Mechanical Engineering Training Simulators

Basic instruction set for the 31 classroom simulators

Istec Training · compiled 30 October 2026 · illustrative training models, not design tools

1. Getting started

Every simulator is a standalone web page. Open the launch centre by double-clicking index.html in the "Mech engineering Training" folder (or run "Start simulators.cmd" on Windows), then pick a simulator. Each simulator can also be opened directly from its own folder under simulators/ by double-clicking its index.html.

The same simulators run online, with nothing to install, at https://chrispienaarthree.github.io/mech-engineering-simulators/ (phones and tablets included).

No server, installation or internet connection is needed for the local copy. Use Google Chrome or Microsoft Edge. On a projector, press the Theme button for a light or dark palette.

Every simulator has the same layout: a rig drawing at the top left with the machine and its instruments; a row of readouts underneath; a control panel with sliders, selectors, action buttons and ready-made scenarios; one to three charts on the right; and a drawer at the bottom with tabs for the Data log, Learning tasks and Behind the numbers.

Record reading stores the present state as a row in the Data log. The log can be copied or downloaded as a CSV file for a spreadsheet. Learning tasks are checked automatically as you work: each turns to "done" when the readings or events it asks for have happened. Behind the numbers gives the equations, the code clauses and the assumptions.

The Guide button on each page repeats the basic steps for that simulator. Space normally starts or pauses the simulation, R records a reading and Esc closes the guide.

Numbers marked [Illustrative data] are typical values chosen for teaching. The models reproduce the physics and the trends faithfully but are not substitutes for manufacturer data or design codes. Code clauses in the tank and pipeline simulators follow the editions named on each page: check them against the edition in force for a real project.

2. The simulators at a glance

No.	Simulator	Topic
01	Hydraulic Loop Bench	Fluid mechanics · centrifugal pumps · pipe friction · control valves · NPSH
02	Restriction Board	Bernoulli · friction · minor losses · flow measurement · system curves
03	Instrumented Pump Trainer	Pump testing · H–Q, efficiency, system and affinity curves · series and parallel operation
04	Positive-Displacement Pump Skid	PD pumps · relief protection · pulsation · back-pressure · calibration
05	Cavitation Trainer	NPSH · cavitation inception and breakdown · vibration and acoustic diagnosis · impeller damage mechanisms
06	Compressor Map Simulator	Centrifugal compressors · surge and choke · anti-surge control · recycle valve dynamics
07	Operating-Point Board	Pump and system curves · operating point · affinity laws · POR/AOR · parallel pumps · impeller trim

08	Hydraulic Power Demonstrator	Hydraulic power · F = p·A · v = Q/A · rod-side areas · relief valve · meter-in / meter-out · power and heat
09	Derrick Crane Demonstrator	Lifting equipment · stiff-leg derrick statics · topping lift, boom and mast forces · reeving · anchors and overturning · load charts
10	Loads on Beams Demonstrator	Beam bending · reactions · shear force and bending moment diagrams · deflection · section properties · stress
11	Mass-Spring-Damper Demonstrator	Vibration fundamentals · natural frequency · damping ratio and log decrement · resonance · frequency response · transmissibility and isolation
12	Planetary Gear Demonstrator	Epicyclic gear trains · Willis equation · speed ratios and reversal · differential action · torque balance · assembly conditions
13	Bearing and Lubrication Rig	Tribology · Stribeck curve · journal bearing film and friction · rolling bearing life, friction, defect frequencies
14	Heat Exchanger Trainer	Heat transfer · ε-NTU and LMTD · counterflow against parallel flow · fouling · pinch and approach
15	Belt and Chain Drive Bench	Power transmission · Euler–Eytelwein traction · installation tension · creep and slip · belt selection · timing belts · chain chordal action
16	Rotating Machinery Fault Simulator	Condition monitoring · vibration analysis · FFT, waveform, phase, envelope · fault diagnosis
17	Brake and Clutch Demonstrator	Friction devices · disc and drum brakes · plate clutch · torque capacity · energy and temperature rise · fade · clutch slip time
18	Fatigue and Stress Concentration Bench	Fatigue · stress concentration Kt and Kf · notch sensitivity · Marin factors · S-N curve · Goodman diagram · Miner damage · Paris crack growth
19	Bolted Joint and Torque Bench	Bolted joints · bolt and member stiffness · joint diagram · torque-tension and its scatter · embedding · separation · proof load · bolt fatigue
20	Steam and Gas Turbine Cycle Board	Thermodynamic cycles · Rankine with superheat, reheat and condenser vacuum · Brayton with pressure ratio, turbine inlet temperature and regenerator · combined cycle · T-s diagrams
21	Control Valve and PID Loop Bench	Process control · control valve characteristics and authority · actuator lag and stiction · PID with anti-windup · step-test identification · tuning rules
22	Refrigeration Cycle Trainer	Vapour-compression refrigeration · p-h diagram · COP · superheat and subcooling · compressor volumetric efficiency · cold-room dynamics and thermostat cycling
23	Rotor Balancing and Critical Speed Rig	Rotor dynamics · Jeffcott rotor · critical speed and phase · damping · run-up · single-plane influence-coefficient balancing · ISO balance grade
24	Shaft Alignment Bench	Shaft alignment · offset and angularity · rim-and-face and reverse-dial readings · bracket sag · shims and moves · thermal growth · soft foot · tolerances by speed · vibration from misalignment
25	Gearbox Tooth Loads and Efficiency Bench	Gear design · tooth forces · Lewis bending and Hertz contact stress · contact ratio · dynamic factor · mesh, churning, bearing and seal losses · oil temperature · pitting and scuffing
26	API 650 Tank Design Demonstrator	Storage tank design · API 650 one-foot method · allowable stresses and materials · corrosion allowance · plate schedule · bottom and annular plates · roofs · wind girders · overturning and anchorage · steel mass against diameter · hydrotest
27	API 620 Tank Design Demonstrator	Low-pressure storage tanks · API 620 membrane design · unit forces T1 and T2 · dome, torispherical and cone roofs · knuckle compression · compression-ring region · buckling allowable · uplift and anchor bolts · Annex R and Q cold service · vapour space, vent and hydro-pneumatic test
28	Gas Pipeline Route and Wall Thickness Designer	Gas transmission pipelines · ASME B31.8 · location class by building count · design factors and exceptions · Barlow design formula with F,

		E, T · standard walls and handling D/t · test pressure · class location change of an existing line · potential impact radius and HCAs
29	Pipeline Hydrostatic Test Bench	Gas pipeline strength and leak test · ASME B31.8 841.3 and Appendix N · test sections from the elevation profile · pressure–volume plot · yield detection · 2 h hold · temperature against leak · MAOP from the test
30	Pipeline Corrosion and Integrity Assessor	Gas pipeline integrity · cathodic protection survey (on and instant-off, IR drop) · Appendix K / NACE RP0169 criteria · coating and soil · stray current · in-line inspection features · ASME B31.8 Appendix L (original B31G) and modified B31G · repairs per 851.44 · MAOP reduction · remaining life
31	Pressure Reduction Station Simulator	Gas pressure control and overpressure protection · ASME B31.8 845 and 841.126 · worker and monitor regulators · slam-shut · relief valve sizing · MAOP + 4 % / + 10 % / 75 % SMYS · sensing lines · Joule–Thomson cooling and freezing · outlet velocity

3. Suggested classroom sequence

Fluids and pumps: 01 Hydraulic Loop, 02 Restriction, 03 Instrumented Pump, 07 Operating-Point, 05 Cavitation, 04 Positive-Displacement Pump Skid, 06 Compressor Map.

Machine elements and structures: 10 Loads on Beams, 18 Fatigue and Stress Concentration, 19 Bolted Joint and Torque, 09 Derrick Crane, 08 Hydraulic Power, 15 Belt and Chain Drive, 12 Planetary Gear, 25 Gearbox Tooth Loads and Efficiency, 13 Bearing and Lubrication, 17 Brake and Clutch.

Dynamics and condition monitoring: 11 Mass-Spring-Damper, 23 Rotor Balancing and Critical Speed, 24 Shaft Alignment, 16 Rotating Machinery Fault.

Thermal and control: 14 Heat Exchanger, 20 Steam and Gas Turbine Cycle, 22 Refrigeration Cycle, 21 Control Valve and PID Loop.

Static equipment and storage: 26 API 650 Tank Design, 27 API 620 Tank Design.

Gas pipelines (ASME B31.8): 28 Gas Pipeline Route and Wall Thickness, 29 Pipeline Hydrostatic Test, 30 Pipeline Corrosion and Integrity, 31 Pressure Reduction Station.

4. Instructions for each simulator

01 · Hydraulic Loop Bench

Folder: simulators/hydraulic-loop/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/hydraulic-loop/index.html

Topic: Fluid mechanics · centrifugal pumps · pipe friction · control valves · NPSH

What it does: Transparent recirculating loop with a variable-speed centrifugal pump, pressure gauges, flowmeter, interchangeable test-section pipe (DN25–DN50), control valve and differential-pressure tapping points. Learners change speed, valve position and pipe resistance and watch the operating point move on the H–Q diagram.

How to use it

1. Start the pump with the speed slider. Watch the impeller, the flow in the clear pipe and the gauges respond. The motor ramps rather than jumping.

2. Throttle the valve and watch the system curve steepen and the operating point slide up the pump curve to lower flow and higher head.

3. Swap the pipe to a smaller bore: the same effect as throttling, but you cannot undo it from the control room.

4. Reduce speed instead of throttling to reach the same flow and compare the electrical power. That difference is the whole argument for variable-speed drives.

5. Break it on purpose. Shut the valve with the pump running (dead-head), heat the fluid and foul the strainer (cavitation), or open everything with DN50 fitted (run-out and overload). Every failure is shown, not just the success.

6. Record points or auto-sweep to build a table you can copy into a spreadsheet.

Keys: ↑ ↓ speed · ←→ valve · R record · S sweep · Esc close

Controls

Motor speed · Control valve opening · Test-section pipe · Test-section length · Fluid temperature · Suction strainer fouling · Reservoir level above pump

Action buttons: Record point · Auto-sweep valve · Reset rig

Readouts

Charts: Head – flow diagram

Learning tasks (checked automatically)

1. **Start the loop.** Run the motor at 100 % with the valve half open and confirm flow on the meter.

2. **Find the best-efficiency point.** At 100 % speed, adjust the valve until the efficiency readout is within 2 % of its maximum. Note the flow and head.

3. **Same flow, two ways.** Deliver 12 m^3/h (±0.5) by throttling at 100 % speed and record it. Then open the valve fully and reduce speed until you get 12 m^3/h again; record that too. Compare electrical power.

4. **Swap the test section.** With speed 100 % and the valve fully open, record a point with DN50 fitted, then with DN25. Watch the operating point slide up the curve and ΔP across the test section climb.

5. **Make it cavitate.** Heat the fluid and foul the strainer with the pump running until the NPSH margin goes negative. Watch the head collapse on the chart and the bubbles at the impeller eye.

6. **Dead-head the pump.** Shut the valve with the pump running for a few seconds. No flow, full head, and the casing starts to heat because all the power goes into the trapped fluid.

7. **Run it off the end of the curve.** Fit DN50, open the valve fully at 100 % speed. The pump runs out to high flow and the motor is overloaded — the other end of the failure envelope.

02 · Restriction Board

Folder: simulators/restriction-board/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/restriction-board/index.html

Topic: Bernoulli · friction · minor losses · flow measurement · system curves

What it does: Interchangeable restriction board with a six-tap piezometer bank: straight pipe (three surface roughnesses), orifice, venturi, sudden expansion, sudden contraction, elbows (1–4, three types) and gate/globe/ball valve. Learners read static and energy lines along the line, ΔP across the element, loss coefficient K, meter-inferred flow against the reference flowmeter, and trace ΔP–Q element curves.

How to use it

1. Pick an element from the board. Its options appear beside it: throat ratio for the meters, downstream or upstream bore for the steps, number and type of elbows, opening for the valve.

2. Set the flow with the supply pump and the downstream throttle. The piezometer tubes rise and fall with the pressure at each tap; the green line is the energy line.

3. Read ΔP between T2 and T4 across the element, and the permanent loss between T1 and T6. For the meters, the inferred flow is compared with the reference flowmeter.

4. Record or auto-sweep to trace the element's ΔP–Q curve on the chart.

5. Break it. Push a venturi hard with warm fluid to cavitate the throat; run an orifice at very low flow to see its coefficient drift; block a tap and see what a stale reading does to the profile.

Keys: ↑ ↓ pump speed · ←→ throttle · 1–7 element · R record · S sweep · Esc close

Controls

Fitted element · Supply pump speed · Downstream throttle · Fluid temperature · Simulate a blocked tapping (T3)

Action buttons: Record point · Auto-sweep flow · Reset

Readouts

Charts: Element curve · ΔP against flow

Learning tasks (checked automatically)

1. **Bernoulli at the venturi throat.** Fit the venturi and run at least 8 m^3/h. Record a point. Check in the tapping-point table that the static head at T3 falls while the total head barely changes — that is Bernoulli, not a loss.

2. **Pressure recovery in an expansion.** Fit the sudden expansion and run at least 8 m^3/h. The T4 column stands higher than T2: static pressure rises as the fluid slows, even though energy is lost. Record it.

3. **Friction scales with Q^2.** Straight pipe: record a point, then double the flow (within 10 %) and record again. Compare the ΔP ratio with 4.

4. **Minor losses add up.** Record one elbow and then four elbows of the same type at the same flow (within 5 %). The ΔP ratio should be close to 4; the K readout tells you the coefficient per elbow.

5. **Calibrate the orifice.** Fit the orifice, run at least 8 m^3/h and adjust your Cd until the meter-inferred flow agrees with the reference flowmeter within 1 %.

6. **Trace an element curve.** Auto-sweep any element. Six or more points trace the ΔP–Q parabola: this is the element's contribution to a system curve.

7. **Cavitate the venturi throat.** Venturi with β 0.5, fluid at 80 °C, everything open at 100 % speed. When the throat pressure reaches vapour pressure the reading stops following the flow.

03 · Instrumented Pump Trainer

Folder: simulators/pump-trainer/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/pump-trainer/index.html

Topic: Pump testing · H–Q, efficiency, system and affinity curves · series and parallel operation

What it does: Closed loop with two transparent pumps and a full instrument board: suction and discharge pressure, flow, rpm, motor current and power, bearing vibration and temperature, fluid temperature. Adjustable speed per pump, discharge and suction throttling, tank level, suction restriction, and single / parallel / series arrangement. Learners construct the H–Q, efficiency, system, affinity and series/parallel curves from recorded points; model curves stay hidden until revealed.

How to use it

1. Curves are built, not shown. The chart starts with only your recorded points. Vary the discharge throttle at fixed speed and record to build the H–Q and efficiency curves; vary speed at fixed throttle to build the system curve; repeat at other speeds for affinity; switch to parallel or series and repeat.

2. Reveal model curves only to check your construction.

3. Sweep discharge throttle records nine points automatically at the current speed and arrangement.

4. Instruments: every reading a real board would give — suction and discharge pressure, flow, rpm, motor current and power, bearing vibration and temperature, fluid temperature. Head and efficiencies are derived and shown in the log so learners can check their own arithmetic.

5. Thermal ×20 speeds up bearing and fluid heating so a dead-head or a long run shows within a lesson.

Keys: ↑ ↓ P1 speed · ←→ discharge throttle · 1 2 3 single / parallel / series · R record · S sweep · Esc close

Controls

Reveal model curves (pump, system, affinity, series/parallel) · Plot efficiency of recorded points (right axis) · Colour points by · P1 speed · P2 speed · Discharge throttle · Suction throttle · Tank level above pump eye · Suction restriction (strainer fouling) · Fluid

Action buttons: Record point · Sweep discharge throttle · Reset rig

Readouts

Charts: Curve builder

Learning tasks (checked automatically)

1. **Construct the H–Q curve.** Single pump at 100 %. Record at least six points across the discharge-throttle range (shut to open). Plot head against flow.

2. **Construct the efficiency curve and find the BEP.** From the same points, compute ρgQH / P_shaft (or read the derived column) and find the flow at peak efficiency.

3. **Construct the system curve.** Fix the discharge throttle (say 60 %) and record points at four or more speeds. These points lie on the system curve, not on a pump curve.

4. **Affinity-law curves.** Record H–Q points at 100 %, 80 % and 60 % speed. Check that shut-off head scales with speed squared and run-out flow with speed.

5. **Parallel curve.** Parallel, both pumps at 100 %. Record at least five points across the throttle range and compare with the single-pump curve: flow adds at equal head.

6. **Series curve.** Series, both pumps at 100 %. Record at least five points: head adds at equal flow.

7. **Predict the operating point.** From your H–Q curve and your system curve, predict the flow at 100 % speed with the throttle at 60 %, then set the rig to that state and record. Within 5 %?

8. **Mismatched pumps in parallel.** Parallel with P2 at 70 % and P1 at 100 %, throttle 30 %. Watch P2's flow: a slower pump against the header pressure of a faster one can be pushed to shut-off.

04 · Positive-Displacement Pump Skid

Folder: simulators/pd-pump-skid/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/pd-pump-skid/index.html

Topic: PD pumps · relief protection · pulsation · back-pressure · calibration

What it does: Small diaphragm or gear pump skid with transparent suction and discharge, gas-charged pulsation dampener, discharge gauge, adjustable relief valve, back-pressure valve and a 500 mL calibration column. The centrepiece: close the discharge progressively and compare the PD pump's rising pressure with a centrifugal pump's flat shut-off — then isolate the relief valve and watch why it is mandatory.

How to use it

1. Start the pump and watch the pulses travel down the transparent discharge line. Switch between diaphragm and gear to compare pulsation.

2. Close the discharge valve progressively — by hand or with the demonstration button — and watch the PD pressure climb while the centrifugal trace flattens at its shut-off head. The relief valve lifts at its set pressure and returns flow to the tank.

3. Isolate the relief valve and close the discharge again: the hose ruptures or the motor stalls. Reset to repair.

4. Dampener: isolate it, or set its charge far from line pressure, and read the ripple on the trace and the gauge.

5. Calibration column: draw from the column for a timed period and compare measured flow with the theoretical displacement.

Keys: ←→ discharge valve · ↑ ↓ speed · D demonstration · R record · C calibration test · Esc close

Controls

Pump speed · Stroke length (diaphragm) · Discharge valve opening · Relief valve set pressure · Relief valve isolated (never do this on a real skid) · Back-pressure valve setting · Dampener gas charge · Dampener isolated · Glycerine-filled gauge · Suction valve opening

Action buttons: Run the closure demonstration · Calibration test: draw from column · Record point · Reset · Start / stop test

Readouts

Charts: Close the discharge progressively · centrifugal against PD · Discharge pressure · last 4 s

Learning tasks (checked automatically)

1. **Calibrate the pump with the column.** Run at 100 % speed with the discharge open; draw from the column for at least 20 s; compute volumetric efficiency. Repeat at 20 % opening and note the effect of pressure on slip.

2. **Close the discharge with the relief in service.** Record points at 100, 60, 40, 20 and 0 % opening. Flow stays almost constant until the relief lifts; the centrifugal trace flattens at its shut-off head instead.

3. **See the relief valve do its job.** With the discharge shut, all the pump flow returns to the tank through the relief valve at set pressure. Compare the relief flow reading with the theoretical displacement.

4. **Isolate the relief valve and close the discharge.** The pressure climbs past the hose rating and the hose ruptures, or the motor stalls first if you set the relief high. This is why PD relief protection is mandatory. Reset to repair.

5. **Measure what the dampener does.** Record ripple with the dampener in service at about 80 % of line pressure, then isolated, then charged far above line pressure.

6. **Back-pressure valve and metering accuracy.** With the discharge fully open and the BPV at 0 bar, then at 2 bar, compare delivered flow — the pump meters against a positive head either way, but the BPV keeps the check valves seating cleanly on a low-resistance line.

7. **Diaphragm against gear.** Repeat the closure with the gear pump: same relief logic, far lower pulsation, higher slip at pressure.

05 · Cavitation Trainer

Folder: simulators/cavitation-trainer/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/cavitation-trainer/index.html

Topic: NPSH · cavitation inception and breakdown · vibration and acoustic diagnosis · impeller damage mechanisms

What it does: Transparent pump and instrumented suction line: suction throttle, vacuum gauge, temperature, flowmeter, accelerometer (waveform and spectrum) and microphone (audible synthesised sound). Learners reduce NPSHa by throttling, heating or suction lift and see, hear and measure cavitation while the head deteriorates; run an NPSH test to find NPSH3; then diagnose four damaged impellers (cavitation, erosion, corrosion, suction recirculation) from the evidence.

How to use it

1. Enable sound (top right) so the pump hum and the cavitation crackle play through the speakers.

2. Set a test flow with the discharge valve, then reduce NPSHa progressively: close the suction throttle, warm the fluid, or lower the liquid level.

3. Watch everything at once: the vacuum gauge falls, bubbles appear at the impeller eye, the microphone level and accelerometer RMS rise, and finally the head and flow drop. The time strip shows the order in which they happen.

4. Record points as you go, or auto-run the test, to build the classic head-versus-NPSHa curve and find NPSH3.

5. Diagnose the impellers in the drawer: four samples, four mechanisms, evidence cards, feedback.

Keys: ←→ suction throttle · ↑ ↓ temperature · R record · S auto-run · M sound · Esc close

Controls

Suction throttle opening · Fluid temperature · Liquid level relative to pump eye · Pump speed · Discharge valve (sets the test flow)

Action buttons: Record NPSH point · Auto-run NPSH test · Reset

Readouts

Charts: NPSH test · head against available NPSH · Last 40 s · head, vibration and sound together

Learning tasks (checked automatically)

1. **Hear it before you see it in the head.** Enable sound. With the discharge valve at 70 %, close the suction throttle slowly. Note the throttle position where the crackle and the accelerometer floor start rising — the head has not moved yet.

2. **Find NPSH3.** Record points as you reduce NPSHa at a fixed discharge-valve setting until the head has dropped 3 %. Read NPSH3 off the chart and compare with NPSHr at that flow.

3. **Temperature alone can do it.** Open the suction throttle fully, then raise the fluid temperature until cavitation develops. Nothing mechanical changed — only the vapour pressure.

4. **Suction lift.** Cool the fluid, open the throttle, and lower the tank level below the pump eye until the margin goes negative. Read the vacuum gauge as it happens.

5. **Recover it.** From developed cavitation, restore a margin above 2.5 m by opening the throttle, cooling the fluid or raising the level. Watch the order in which sound, vibration and head come back.

6. **Diagnose all four impellers.** In the Damaged impellers tab, name the mechanism for every sample from the location and texture of the damage and the service history.

06 · Compressor Map Simulator

Folder: simulators/compressor-map/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/compressor-map/index.html

Topic: Centrifugal compressors · surge and choke · anti-surge control · recycle valve dynamics

What it does: Single-stage centrifugal process compressor on a live map: speed lines, surge line, surge control line, choke line, system line and the operating point with its trail. A P&ID shows the compressor, aftercooler, check valve, process valve and the anti-surge recycle valve. Learners change suction pressure, discharge header pressure, speed, process demand and the recycle valve stroke time, and watch normal → approaching surge → surge → recovery on pressure, flow, valve and vibration traces. Surge cycles come from a Greitzer-type plenum model; the anti-surge controller (PI, quick-open, surge counter) can be put in manual.

How to use it

1. Start the compressor. It runs up with the recycle valve open and the anti-surge controller (ASC) closes the valve as the process takes the flow. Watch the operating point on the map settle right of the surge control line.

2. Push it toward surge with any of the four sliders: suction pressure down, header pressure up, speed down, process demand down. The status pill changes from normal to approaching surge as the point enters the control band, and to surge when the flow reverses.

3. Recycle valve response sets how fast the anti-surge valve can move. Try the same scenario at 2 s and at 15 s. Put the ASC in Manual to protect the machine yourself, or to let it surge.

4. Traces show discharge and suction pressure, inlet flow with the surge control flow, recycle valve position and shaft vibration, with a phase band along the top. A vibration trip or the surge counter stops the machine.

5. Scenarios under the controls ramp the process for you; "Surge sequence" plays normal → approaching → surge → recovery in about a minute.

Keys: ↑ ↓ speed · ←→ process demand · S start / stop · M auto / manual · R record · Esc close

Controls

Suction pressure Ps · Discharge header pressure · Speed setpoint · Process demand (process valve) · Recycle valve response — full stroke · Surge control margin · Anti-surge controller · Manual opening

Action buttons: Start · Stop · Reset trip · Record point

Readouts

Inlet flow · Mass flow · Pressure ratio · Surge margin · Recycle valve · Shaft vibration · Speed · Shaft power

Charts: Compressor map · pressure ratio against inlet flow · Traces · last 60 s

Ready-made scenarios and presets

Design point · Surge sequence (slow valve) · Close the process valve · Header pressure creeps up · Speed cut to 85 % · Suction pressure falls · ASC off — recover by hand

Learning tasks (checked automatically)

1. **Start and load the compressor.** Start, let it run up with the recycle open, and watch the ASC close the valve as the process takes the flow. Where does the point settle relative to the control line?

2. **Enter the control band.** Move the point into the band between the surge control line and the surge line with any of the four sliders. Watch the vibration begin to rise before anything else does.

3. **Make it surge.** Cross the surge line (ASC in manual with the valve shut, or a slow valve) and watch the map, the pressure trace and the vibration trace during a surge cycle.

4. **Recover it by hand.** With the ASC in manual, open the recycle valve until the oscillation stops and the point sits right of the control line. Note how much recycle was needed.

5. **Slow valve loses the race.** ASC auto, valve stroke 10 s or more, then close the process slowly. Count the surge cycles before the recycle catches up.

6. **Fast valve catches it.** Same disturbance with the stroke at 2 s or less. The point enters the band, the valve opens and the point never crosses the line.

7. **Suction pressure alone.** Change nothing downstream; drop the suction pressure to 3.8 bar a or less and explain why the point still moves toward surge.

8. **Find the stonewall.** Open the process valve fully and lower the header until the point runs out to the choke line. What happens to flow when you ask for more?

9. **Let the protection trip it.** Leave a surge unattended (manual, valve shut) or set a very slow valve: the surge counter or the vibration trip stops the machine.

07 · Operating-Point Board

Folder: simulators/operating-point-board/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/operating-point-board/index.html

Topic: Pump and system curves · operating point · affinity laws · POR/AOR · parallel pumps · impeller trim

What it does: Digital twin of the magnetic H–Q board (print set in operating-point-board/print). Teams drag pump-curve pieces (speed, trimmed impeller, two in parallel), system parabolas that slide to the static head, POR/AOR bars and BEP dots onto the chart, mark a predicted operating point and check it against the true intersection. Seven scenario cards: baseline, strainer fouls 40 %, tank level falls 5 m, throttle to 18 m^3/h, VFD to 1 200 rpm, second pump in parallel, impeller trim. Projector mode hides the pieces so the screen only reveals answers for the physical board.

How to use it

1. Pick a scenario card on the right. It says which pieces you need; those buttons in the tray are outlined in brass.

2. Add pieces from the tray and drag them onto the chart. Pump curves and bars snap when you register them to the grid (Q = 0 end, white ticks). System parabolas snap to Q = 0 and slide up and down: their lower end sits on the static head.

3. Predict: drag the red ? marker to where the curves cross. Its Q and H show on the right. Press Check: the true curves and the true point appear and the result is logged for the team.

4. Projector mode hides the digital pieces so the screen only reveals the answer for teams working on the magnetic board.

Keys: 0 A–F scenario · Enter check · R reveal · Esc close

Controls

Team

Action buttons: Check prediction · Reveal answer · Hide answer

Readouts

Predicted Q · Predicted H · System piece · Pump piece

Learning tasks (checked automatically)

1. **Commission the board.** Card 0: register P1, S2 at 10 m and W1, put the BEP dot on the curve, and hit the operating point.

2. **Strainer fouls 40 %.** Card A: swap S2 for S3, keep the static head, hit the new point.

3. **Tank level falls 5 m.** Card B: slide S2 up to 15 m, hit the new point.

4. **Throttle to 18 m^3/h.** Card C: choose the system piece that crosses P1 at 18 m^3/h, hit the point.

5. **VFD to 1 200 rpm.** Card D: place P3 with its own bar W3 and BEP, hit the point.

6. **Second pump in parallel.** Card E: place P8 and W8, hit the total flow point.

7. **Trim for 18 m^3/h.** Card F: place P7 and W7, hit the point.

8. **All four routes compared.** Cards A to D each checked at least once (hit or close): which change cost the most flow, and which the least power?

08 · Hydraulic Power Demonstrator

Folder: simulators/hydraulic-power-demonstrator/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/hydraulic-power-demonstrator/index.html

Topic: Hydraulic power · F = p·A · v = Q/A · rod-side areas · relief valve · meter-in / meter-out · power and heat

What it does: 4 kW power pack (variable-speed gear pump, relief valve, 4/3 directional valve, meter-in and meter-out needle valves) driving a vertical cylinder chosen from a rack of bores (25–80 mm) and rods, with a load cell on the rod and gauges on the pack and both ports. Loads: dead weight, spring, friction brake. Learners change the cylinder dimensions and read force and speed, stall it at the relief setting, intensify the rod side by closing the meter-out valve, waste power through the meter-in valve and watch the tank heat, overload the motor, and control an overrunning load. Trade-off chart of force and speed for every bore.

How to use it

1. Pick a cylinder from the rack (bore and rod). The drawing scales with it and the piston-face inset shows the two areas.

2. Start the pump, set the relief valve, and hold Extend or Retract. Read P1 on the pack, P2 and P3 at the ports, and the force at the rod on the load cell. Compare it with p·A.

3. Change the load: a dead weight, a spring that stiffens with stroke, or a brake that never yields. Stall the cylinder and watch the pressure sit at the relief setting.

4. Valves: meter-in wastes power at the relief valve; meter-out controls overrunning loads but intensifies rod-side pressure when closed.

5. Record readings at several bores and let the tasks check them.

Keys: ↑ hold to extend · ↓ hold to retract · S start / stop pump · R record · 1–6 bore · Esc close

Controls

Cylinder bore · Piston rod · Load on the platform · Mass · Relief valve setting · Pump speed (VSD) · Meter-in valve (supply) · Meter-out valve (return) · Closed-centre directional valve (pump dead-headed in neutral) · Directional valve

Action buttons: Start pump · Stop pump · Reset trip / repair · Record reading

Readouts

P1 pump · P2 cap side · P3 rod side · Force at rod (load cell) · Rod speed · Areas · Pump flow · Motor power · Wasted as heat · Oil temperature

Charts: Force and speed against bore · at the current relief setting and pump flow · Traces · last 40 s

Ready-made scenarios and presets

Small bore, fast · Big bore, strong · Stall test (brake) · Lift 2 000 kg · Meter-out closed · Wasteful meter-in · Overrunning load

Learning tasks (checked automatically)

1. **Force = pressure × area.** Fit the brake (or a load the cylinder cannot move), stall it, and record the load-cell force at three different bores with the same relief setting. Plot force against area.

2. **Speed = flow ÷ area.** Record the extending speed at three bores with the same pump speed and both valves open.

3. **Rod side: less force, more speed.** For one cylinder, record a moving extend and a moving retract reading. Compare the speeds and the p·A forces with the area ratio.

4. **Stall it.** Raise the load until the cylinder stops. Where does P1 sit, and what does the load cell read compared with p·A?

5. **Intensify the rod side.** Extend against a load with the meter-out valve almost closed. Watch P3 climb past the relief setting.

6. **Where the power goes.** Throttle the meter-in valve to half speed or less while lifting. Read the wasted power and watch the tank temperature.

7. **Overload the motor.** High relief setting, full pump speed, big load: push the motor past 4 kW until it trips.

8. **Size the cylinder.** Lift 2 000 kg to full stroke with the relief valve at 150 bar or less, using the smallest bore that can do it.

9. **Overrunning load.** Lower a heavy weight with the meter-out valve wide open, then hold it back with the valve.

09 · Derrick Crane Demonstrator

Folder: simulators/derrick-crane/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/derrick-crane/index.html

Topic: Lifting equipment · stiff-leg derrick statics · topping lift, boom and mast forces · reeving · anchors and overturning · load charts

What it does: Stiff-leg derrick with every dimension adjustable: mast height, boom length and foot height, sill reach, leg spread, boom mass, hoist and topping reeving, rope sizes, hook block, anchor hold-down, strut capacities, winch pull. Luff, slew and hoist a load off the ground and read the topping lift, boom and mast compression, both leg forces, anchor uplift and base reaction, coloured by utilisation on the elevation and plan drawings. The load chart (rated load against radius, coloured by the governing limit) is rebuilt from the same solver whenever a dimension changes.

How to use it

1. Set the rig with the dimension sliders: mast height, boom length and foot height, sill reach, leg spread. Change rigging: parts of line, rope sizes, block mass, anchor hold-down, strut capacities, winch pull.

2. Operate it: hoist to lift the load off the ground, luff the boom out and in, slew left and right. Members recolour with utilisation; the readouts give every force.

3. Load chart shows the rated load at every radius for the rig as it is now, coloured by what governs, with the current lift as a red dot. Utilisation bars show how close each part is to its limit.

4. Record readings to compare reeving, mast height, boom angle, boom length and slew. The tasks check the comparisons.

Keys: Q/A luff in/out · Z/X slew · ↑ / ↓ hoist · R record · Esc close

Controls

Mast height · Boom length · Boom foot height above base · Sill reach (leg anchors from mast) · Leg spread (half-angle from the rear) · Boom mass · Hoist reeving (parts of line) · Topping-lift reeving (parts) · Hoist rope · Topping rope · Hook block mass · Anchor hold-down (each) · Boom strut capacity at 20 m · Leg strut capacity at 20 m · Winch line pull (hoist and topping) · Load on the ground · Boom angle · Slew (0 = away from the legs)

Action buttons: Record reading · Land the load

Readouts

Radius · Load on hook · Topping lift · Hoist rope · Boom compression · Mast compression · Leg 1 (left) · Leg 2 (right) · Anchor uplift · Base reaction

Charts: Load chart · rated load against radius for this rig · Member utilisation · % of capacity (ropes at safety factor 5)

Ready-made scenarios and presets

Standard 24 m / 30 m rig · Reach out to the limit · Tall mast, same lift · Narrow legs, slewed · Light anchors · Single-part hoist · Long boom

Learning tasks (checked automatically)

1. **Find the radius limit.** Lift 10 t or more and luff out until something reaches 100 %. Which limit governs, and at what radius?

2. **Reeving halves the rope pull.** Record the same lift with two different hoist reevings and compare the hoist rope tension.

3. **Topping lift against boom angle.** Record the same lift at three boom angles. Plot topping-lift tension against radius.

4. **A taller mast.** Record the same lift (same boom, angle, load) with two mast heights. Where does the topping tension go, and the mast compression?

5. **Slew toward a leg.** With a load on, slew until one leg carries more than twice the other. Keep going: which leg goes into compression?

6. **Lift an anchor.** Reduce the hold-down or increase the load until an anchor lifts and the derrick would overturn.

7. **Longer boom, same radius.** Record the same load at the same radius (±1 m) with two boom lengths. Compare boom compression with its capacity.

8. **Design a 10 t lift at 20 m.** Choose dimensions and rigging so that 10 t at 20 m radius or more has every member below 90 %.

9. **Two-block it.** Hoist until the block meets the boom tip. What does the hoist rope see if the winch keeps pulling?

10 · Loads on Beams Demonstrator

Folder: simulators/beam-loads/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/beam-loads/index.html

Topic: Beam bending · reactions · shear force and bending moment diagrams · deflection · section properties · stress

What it does: Beam bench with adjustable span, support arrangement (simply supported with movable supports and overhangs, cantilever, propped cantilever, fixed ends), draggable point loads, distributed loads and couples, parametric sections (rectangle, I, hollow rectangular, round, hollow round) and three materials. A beam finite-element solver gives reactions and deflection for any arrangement; shear and moment diagrams are built by statics. Reads reactions, extremes of shear, moment, stress and deflection, contraflexure count, utilisation against allowable stress and a span/limit. Failure modes: yield, deflection limit, support uplift, mechanism.

How to use it

1. Set the beam: span, support arrangement and support positions. Supports can be dragged along the beam for overhangs.

2. Add loads: point loads, distributed loads and couples from the load list. Drag the red point-load arrows and the support triangles on the drawing.

3. Read reactions on the drawing, the extremes in the readouts, and the shear, moment and deflection diagrams on the right. The deflected shape is drawn exaggerated on the beam.

4. Section and material change stress and deflection but not the moment: compare a rectangle with an I-section, steel with aluminium.

5. Record readings and let the tasks check them against the standard results.

Keys: R record · P add point load · W add distributed load · Esc close

Controls

Span (overall length) · Support arrangement · Include beam self-weight · Support A position · Support B position · Deflection drawing scale · Section · Flange width b · Depth h · Flange thickness t · Material · Deflection limit span/

Action buttons: Record reading · Clear loads

Readouts

Reaction A · Reaction B · Max shear · Max moment · Bending stress · Max deflection · Section · Total load · Contraflexure · Utilisation

Charts: Shear force · bending moment · deflection

Ready-made scenarios and presets

Central point load · Uniform load · Four-point bending · Cantilever tip load · Overhangs, uniform load · Propped cantilever · Fixed both ends · Applied couple · Load on the overhang

Learning tasks (checked automatically)

1. **$wL^2/8$ and $wL/2$.** Simply supported at the ends with one uniform load over the whole span. Record it and check $R = wL/2$ and $M = wL^2/8$.

2. **PL/4.** One point load at mid-span on a simply supported beam. Record it; check $M = PL/4$ and $\delta = PL^3/48EI$.

3. **Cantilever tip load.** Fixed at A, one point load at the tip. Record it; check $M = PL$ and $\delta = PL^3/3EI$.

4. **Best overhang.** Uniform load over the whole beam: slide both supports in until the sagging and hogging moments are equal (about 0.207 L).

5. **Double the depth.** Record the same beam and loads with a section depth h and then 2h. Deflection should drop eight times for a rectangle.

6. **Fixed ends.** Record the full uniform load simply supported and then fixed at both ends: the peak moment falls from $wL^2/8$ to $wL^2/12$ and moves to the supports.

7. **Lift a support.** Put a load on an overhang until the far roller reaction goes negative.

8. **Point of contraflexure.** Record a case where the moment changes sign (propped cantilever, fixed ends, or an overhang).

9. **Size the beam.** 20 kN/m over a 6 m simply supported span: pick a section and material with stress at 60–100 % of allowable and deflection within the limit.

11 · Mass-Spring-Damper Demonstrator

Folder: simulators/mass-spring-damper/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/mass-spring-damper/index.html

Topic: Vibration fundamentals · natural frequency · damping ratio and log decrement · resonance · frequency response · transmissibility and isolation

What it does: Single-degree-of-freedom rig integrated in real time (RK4, adjustable clock): mass plates, spring, damper (or set ζ directly), stops at ±120 mm. Excitation: free release, harmonic force actuator, shaker base, step and impulse. Measured frequency and log decrement from the trace, settled amplitude ratios from a swept sine plotted over the theoretical magnification, transmissibility and phase curves. Time traces with decay envelope, phase portrait, energy bars. Nine checked tasks from measuring f_n to designing an isolator.

How to use it

1. Set the system: mass, spring, damper (or pick ζ directly). The readouts give fn, ζ, fd, time constant and log decrement.

2. Choose the excitation: release from a displacement, drive with a harmonic force, shake the base, apply a step, or tap it with an impulse. Press Start.

3. Read the traces: the envelope and zero crossings give the measured frequency and decrement; the frequency-response chart shows where you are on the curve and the sweep plots measured points over it.

4. Slow the clock to watch fast systems; Record readings to compare cases in the tasks.

Keys: Space start / stop · R record · 1–5 excitation mode · Esc close

Controls

Mass (plates) · Spring stiffness k · Damper c · Clock · Initial displacement · Initial velocity · Force amplitude F_0 · Base amplitude Y · Excitation frequency · Stepped sine 0.2 → 3 fn, dwelling at each step until settled, plotting measured points · Impulse

Action buttons: Start / release · Stop · Record reading · Clear measured points

Readouts

Natural frequency · Damping ratio ζ · Damped frequency · Log decrement δ · Displacement · Frequency ratio r · Magnification · Peak velocity · acceleration

Charts: Time traces · last 10 s · Frequency response · magnification and transmissibility against r = f / fn · Phase portrait · velocity against displacement

Ready-made scenarios and presets

Free decay · Critical damping · Overdamped · Resonance · Sweep the FRF · Isolation r = 3 · Car on a wavy road · Step · Hammer tap

Learning tasks (checked automatically)

1. **Measure the natural frequency.** Release the mass with light damping, let it swing a few cycles and record: the measured frequency should match $f_n = \sqrt{(k/m)}/2\pi$.

2. **Four times the mass, half the frequency.** Record two free decays with the same spring and masses in the ratio 4:1.

3. **Log decrement gives ζ.** Record two free decays with different damper settings; compare measured ζ from the decrement with $c/2\sqrt{(km)}$.

4. **Critical damping.** Set ζ = 1 and release from a displacement: no overshoot and the fastest return.

5. **Overdamped creep.** Set ζ = 2 and release: record it and compare the settling with critical.

6. **Find resonance.** Harmonic force with ζ ≤ 0.1: tune the frequency to r ≈ 1 and watch the amplitude climb toward 1/2ζ.

7. **Sweep the response.** Run a swept sine and collect at least 12 measured points on the frequency-response chart.

8. **Isolate the base.** Base excitation with r > √2: measured transmissibility below 1. Then add damping and see it get worse.

9. **Design an isolator.** A machine forcing at 10 Hz: choose m, k and c so that transmissibility is below 0.2 at 10 Hz and the amplitude stays under 20 mm.

12 · Planetary Gear Demonstrator

Folder: simulators/planetary-gears/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/planetary-gears/index.html

Topic: Epicyclic gear trains · Willis equation · speed ratios and reversal · differential action · torque balance · assembly conditions

What it does: Animated epicyclic gearset with adjustable sun and ring tooth counts and number of planets, drawn with teeth kept in phase so the meshes stay in step. Hold the ring, carrier or sun and drive another member, drive two members as a differential, or lock the train. A marker traces the path of a point on one planet as it spins and orbits. Readouts for every member's speed, direction and torque, ratio, planet spins per orbit, power and mesh frequency; speed nomograph (lever analogy), tabular-method table, assembly and clash checks. Nine checked tasks.

How to use it

1. Choose the gears: sun and ring tooth counts (the planet count of teeth follows), and how many planets. The assembly check tells you whether they fit.

2. Hold one member and drive another, set the speed and torque. Watch the marker on one planet trace its path while the planet spins and orbits. Slow the animation clock to follow the teeth.

3. Read the numbers: every member's speed and torque, the ratio and direction, planet spins per orbit, power and mesh frequency. The nomograph and the tabular method show where they come from.

4. Differential: hold nothing and drive two members; find the pair of speeds that stops the carrier.

Keys: Space run / stop · . step one tooth · R record · 1–3 hold ring / carrier / sun · Esc close

Controls

Sun teeth Ns · Ring teeth Nr · Planets · Input speed · Second input (ring) · Input torque · Efficiency per mesh · Animation clock · Trace the path of a point on one planet

Action buttons: Run · Step one sun tooth · Record reading · Clear path

Readouts

Sun · Carrier (orbit) · Ring · Planet spin · Speed ratio · Planet "days per year" · Power · Mesh frequency

Charts: Speed nomograph · the lever analogy · Tabular method · Willis equation · Speed of every member as the input turns · last revolutions

Ready-made scenarios and presets

Reducer · ring held, sun in · Star · carrier held · Overdrive · sun held, ring in · Differential · two inputs · Stop the carrier · Locked 1 : 1 · High ratio 10 : 1 · Won't assemble

Learning tasks (checked automatically)

1. **Ring held: 1 + Nr/Ns.** Hold the ring, drive the sun. Record it and check the ratio against 1 + Nr/Ns. Which way does the carrier turn?

2. **Carrier held: reversal.** Hold the carrier: the ring turns the other way at −Nr/Ns. Record it.

3. **Sun held: 1 + Ns/Nr.** Hold the sun and drive the ring: the smallest ratio of the three. Record it.

4. **Three ratios, one gearset.** Record all three held-member cases with the same tooth counts.

5. **Stop the carrier with two inputs.** Hold nothing; drive sun and ring so that the carrier stands still: ωs·Ns = −ωr·Nr.

6. **Days per year.** With the ring held, note how many times the planet spins per orbit; change Ns and see it change.

7. **Torque balance.** Record a ring-held case and check Ts + Tc + Tr = 0 and Tc/Ts = −(1 + Nr/Ns).

8. **Make it assemble.** Set four planets with tooth counts that fail the assembly check, then change the ring so they fit.

9. **Ten to one in one stage.** Find tooth counts that give a ratio of 10 or more with the ring held and at least three planets that assemble.

13 · Bearing and Lubrication Rig

Folder: simulators/bearing-lube-rig/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/bearing-lube-rig/index.html

Topic: Tribology · Stribeck curve · journal bearing film and friction · rolling bearing life, friction, defect frequencies

What it does: Two modes. Journal bearing: load, speed, diameter, L/D, clearance ratio, roughness, oil grade and supply; short-bearing solution gives eccentricity, minimum film, λ and regime, friction blended across boundary, mixed and hydrodynamic lubrication, film temperature on a thermal clock with viscosity feedback; Stribeck curve with a speed sweep; wiping and overheating failures. Rolling bearing: six catalogue bearings, radial and axial load, contamination and reliability; equivalent load, L10 and modified life, viscosity ratio, Palmgren friction, temperature, speed limit, minimum load and BPFO/BPFI/BSF/FTF; life-against-load chart.

How to use it

1. Journal mode: set load, speed, geometry and oil, press Start. Watch the journal find its eccentricity, the film thickness and λ, the oil warm up and the point move on the Stribeck curve. The speed sweep traces the curve for you.

2. Rolling mode: pick a bearing, set radial and axial load, speed and lubricant. Read the equivalent load, L10 and modified life, friction torque, temperature and defect frequencies; the chart shows life against load.

3. Failure modes: metal contact and wiping, oil overheating, pressure above the babbitt limit, speed above the bearing limit, too little load to keep the elements rolling.

4. Record readings to compare oils, clearances and loads in the tasks.

Keys: Space start / stop · R record · 1 / 2 mode · Esc close

Controls

Radial load · Shaft speed · Axial load · Journal diameter · Length / diameter · Clearance ratio c/R · Combined roughness · Bearing · Contamination · Reliability · Oil grade · Oil supply / ambient temperature · Oil supply · Thermal clock

Action buttons: Start · Stop · Speed sweep (Stribeck) · Record reading · Replace bearing

Readouts

Sommerfeld number · Eccentricity ratio · Minimum film · Friction coefficient · Power loss · Oil temperature · Bearing pressure · Oil flow · Clearance · Surface speed · Equivalent load P · Basic life L10 · Modified life · Load ratio C/P · Viscosity ratio κ · Friction torque · Bearing temperature · Speed · Defect frequencies · Minimum load check

Charts: Stribeck curve · friction coefficient against μN/P · Traces · last 5 min of thermal time

Ready-made scenarios and presets

Design point · Slow and heavy (boundary) · Thin oil, hot supply · Oil supply lost · Tight clearance · Rolling bearing at rated load · Axial load on a ball bearing · Contaminated grease

Learning tasks (checked automatically)

1. **Cross the Stribeck curve.** Journal mode: record one reading in boundary lubrication (λ < 1) and one with a full film (λ > 3). The speed sweep will take you across.

2. **Heavier oil, thicker film, hotter bearing.** Record the same load and speed with two oil grades. Compare film thickness and film temperature.

3. **Double the load.** Record a journal reading, double the load, record again: film thickness falls, friction coefficient falls, power loss rises.

4. **Clearance.** Record two clearance ratios at the same conditions. Tight clearance gives a thinner film and more heat; loose gives less load capacity.

5. **Lose the oil supply.** Cut the oil supply while running: watch the film temperature run away and the bearing wipe.

6. **The cube law.** Rolling mode: record a bearing at one radial load and at double the load. L10 should fall eight times for a ball bearing.

7. **Axial load on a ball bearing.** Put an axial load on a deep-groove ball bearing until Fa/Fr exceeds e: the equivalent load jumps and life collapses. Record it.

8. **Clean versus dirty.** Record the same rolling bearing with normal cleanliness and heavy contamination: compare the modified life.

9. **Speed and viscosity.** Run a rolling bearing past its speed limit, or with too heavy an oil at high speed, until the temperature alarm shows.

14 · Heat Exchanger Trainer

Folder: simulators/heat-exchanger/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/heat-exchanger/index.html

Topic: Heat transfer · ε-NTU and LMTD · counterflow against parallel flow · fouling · pinch and approach

What it does: Counterflow, parallel, 1-2 shell-and-tube, cross-flow and plate arrangements with water, oil or glycol on each side, adjustable flows, inlet temperatures, area and clean U. ε-NTU rating with film coefficients scaled by flow, fouling growing on a compressed clock and cleanable, a hot-side bypass valve, pressure drops and pumping power. Temperature profiles with the pinch marked, ε-NTU chart of all arrangements at the present Cr, duty and fouling traces. Alarms for boiling, fouling, high pressure drop and impossible temperature cross.

How to use it

1. Pick an arrangement, set both streams (fluid, flow, inlet temperature), the area and the clean U, then Start. Read the duty, effectiveness, NTU, outlet temperatures, LMTD and pressure drops.

2. Profiles show the two streams along the exchanger with the approach and pinch marked; the ε-NTU chart shows every arrangement at the present capacity ratio with your point on it.

3. Fouling builds on the fouling clock and the duty falls; clean the exchanger to recover it. The bypass valve lets you control the hot outlet.

4. Record readings to compare arrangements, flows and sizes in the tasks.

Keys: Space start / stop · C clean · R record · 1–5 arrangement · Esc close

Controls

Fluid · Mass flow · Inlet temperature · Heat-transfer area · Clean U at design flows · Fouling rate · Hot-side bypass (control valve) · Fouling clock

Action buttons: Start · Stop · Clean the exchanger · Record reading

Readouts

Duty Q · Effectiveness ε · Hot out · Cold out · LMTD · Overall U · Fouling · Capacity rates · Pressure drop · Area

Charts: Temperature profile along the exchanger · Effectiveness against NTU · every arrangement at the present Cr · Traces · duty, outlet temperatures and fouling over time

Ready-made scenarios and presets

Oil cooler, counterflow · Same cooler, parallel flow · Balanced flows (Cr = 1) · Cold flow starved · Fouling over 60 days · Plate exchanger, small · Shell-and-tube 1-2 · Push the cold side to boiling

Learning tasks (checked automatically)

1. **Counterflow against parallel flow.** Record the same exchanger and flows in counterflow and in parallel flow. Compare duty and cold outlet.

2. **Make a temperature cross.** In counterflow, get the cold outlet above the hot outlet and record it. Then try the same in parallel flow.

3. **Balanced flows.** Set the flows so the capacity rates match (Cr within 0.95–1.05) and record: parallel temperature lines, lowest ε for the NTU.

4. **Find the pinch.** Starve one stream until the approach temperature falls below 3 K and record it. Which stream limits the duty now?

5. **Double the cold flow.** Record, double the cold flow, record again: duty up, cold outlet down, pressure drop four times.

6. **Foul it, then clean it.** Run with fouling until the duty has fallen 15 %, then clean the exchanger and watch it recover.

7. **Size the oil cooler.** Oil 2 kg/s at 90 °C, water 3 kg/s at 25 °C, counterflow, clean: choose the smallest area that brings the oil out at 50 °C or below.

8. **The F factor.** Record a 1-2 shell-and-tube reading: F below 1 shows the penalty for the partial parallel-flow pass.

9. **Control with the bypass.** Use the hot-side bypass to hold the hot outlet at 55 ± 2 °C while the flows stay put. Record it.

15 · Belt and Chain Drive Bench

Folder: simulators/belt-drive/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/belt-drive/index.html

Topic: Power transmission · Euler–Eytelwein traction · installation tension · creep and slip · belt selection · timing belts · chain chordal action

What it does: V-belt (SPZ–SPC), flat belt, synchronous belt and roller chain on two pulleys with adjustable diameters or teeth, centre distance, belt count, misalignment, driver speed, load torque, installation tension and friction. Wrap angle, tensions, centrifugal tension, capacity against demand, creep below and gross slip above the limit, driven speed, power, efficiency, belts required, shaft load, belt temperature and wear on a clock; tooth jump for timing belts, chordal speed variation, safety factor and lubrication regime for chains. Traction chart with capacity line and slip; chordal pulse chart for chains.

How to use it

1. Choose the drive: V-belt, flat belt, synchronous belt or roller chain, then pulley or sprocket sizes, centre distance, section and number of belts.

2. Run it: set the driver speed and the load torque. Read the tensions on the drawing, the slip, the driven speed and the power. Raise the load until the belt slips, or lower the installation tension.

3. Chart: transmitted tension and slip against load, with the capacity line for the present tension and wrap. For a chain the chart shows the chordal speed pulse over one tooth.

4. Record readings to compare tension, wrap, V against flat and belt count in the tasks.

Keys: Space start / stop · R record · 1–4 drive type · Esc close

Controls

Driver pulley diameter · Driven pulley diameter · Centre distance · Belt section / chain size · Number of belts / strands · Pulley misalignment · Driver speed · Load torque at the driven shaft · Installation tension (of recommended) · Friction coefficient (belt to pulley) · Ambient temperature · Wear clock

Action buttons: Start · Stop · Re-tension / new belt · Record reading

Readouts

Speed ratio · Slip / creep · Belt speed · Wrap on small pulley · Tight side T_1 · Effective tension · Power · Belts required · Shaft / bearing load · Belt temperature

Charts: Traction · transmitted tension against load, with the capacity of this belt and tension · Traces · driven speed, slip and belt temperature

Ready-made scenarios and presets

Design point (SPA × 2) · Load until it slips · Belt too loose · Belt too tight · Big ratio, small wrap · Same drive, flat belt · Timing belt, then jump · Chain, 11-tooth sprocket

Learning tasks (checked automatically)

1. **Load it until it slips.** Raise the load torque until the V-belt passes its capacity: watch the driven speed fall and the belt heat.

2. **Tension sets capacity.** Record the same load at two installation tensions and compare slip and capacity.

3. **Wrap angle.** Record two drives whose small-pulley wrap differs by 20° or more (change the ratio or the centre distance) and compare capacity.

4. **V against flat.** Record the same pulleys, tension setting and load with a V-belt and with a flat belt. Which carries more before slipping?

5. **Select the belts.** For at least 10 kW at the load, choose a section and number of belts so that the belts required equals the belts fitted. Record it.

6. **Over-tensioned.** Set the installation tension to 160 % or more and read the shaft load: the bearings pay for tension the drive does not need.

7. **Tooth jump.** Synchronous belt: lower the tension or raise the load until the belt jumps a tooth.

8. **Chordal action.** Chain: record an 11-tooth driver and one with 19 teeth or more; compare the chordal speed variation.

9. **Belt overheating.** Run a belt in gross slip or with misaligned pulleys until the temperature alarm shows, then read the life factor.

16 · Rotating Machinery Fault Simulator

Folder: simulators/vibration-fault-simulator/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/vibration-fault-simulator/index.html

Topic: Condition monitoring · vibration analysis · FFT, waveform, phase, envelope · fault diagnosis

What it does: Motor-driven shaft rig with interchangeable discs, bearings, coupling, belt drive and gearbox. Learners manufacture unbalance, misalignment, looseness, bent shaft, bearing defects, gear and belt defects and resonance, then measure them with accelerometers and a tachometer: overall level with ISO zone, FFT spectrum in orders or hertz, time waveform, 1× phase across the train, and envelope spectrum. The signal is synthesised from the machine state and analysed for real, so the peaks appear only when the machine produces them. Blind-machine mode hides a fault for diagnosis.

How to use it

1. Manufacture a fault in the panels under the machine: add a mass to the disc, mis-shim the motor, loosen a bearing, fit a damaged bearing, damage a gear or belt, or move the pedestal resonance.

2. Measure it: pick a point and direction, read the overall level and ISO zone, and read the spectrum, waveform, envelope spectrum and the 1× phase table. Everything is computed from a synthesised acceleration signal, so peaks appear only when the machine actually produces them.

3. Order axis shows the spectrum in multiples of running speed, which is how faults are read; switch it off for hertz. Fault-frequency cursors are off by default so learners identify lines themselves.

4. Record readings on a route and export them. Blind machine hides a random fault for diagnosis.

Keys: ↑ ↓ speed · H V A direction · 1–5 point · R record · Esc close

Controls

Belt drive to fan · Gearbox · Second disc · Motor speed · Unbalance mass on disc 1 · at angle from key · Correction / trial mass · Angular misalignment · Parallel offset · Coupling · Bearing 2 hold-down bolt torque · Bearing fit clearance (looseness type C) · Shaft bend (run-out at disc) · Bearing with the defect · Defect · Severity (stage 1 → 4) · Lubrication · Gear defect · Gear defect severity · Belt condition · Belt fault severity · Pedestal natural frequency · Damping ratio ζ · Measurement point · Fmax · orders of running speed · fault-frequency cursors · log amplitude

Action buttons: Bump test (machine stopped) · Record reading

Readouts

Overall 10–1 000 Hz · Peak g · crest

Charts: FFT spectrum · velocity · Waveform · acceleration · Envelope spectrum · 2–6 kHz demod · 1× amplitude and phase across the train

Learning tasks (checked automatically)

1. **Manufacture the 1× peak.** Add at least 15 g to disc 1 at 1 480 rpm and watch 1× rise at bearing 1 H. Note the phase.

2. **Balance it.** Use the tacho phase to place a correction mass. Done when the resultant unbalance is below 20 % of the mass you added.

3. **Angular misalignment across the coupling.** Set at least 3 mm/100 mm angular, then record axial readings at motor DE and bearing 1. The phases differ by about 180°.

4. **Looseness harmonics.** Back the bearing 2 bolts off to 40 % or less and record bearing 2 vertical. Count the harmonics and look for half-order lines.

5. **Find the bearing defect in the envelope.** Fit a stage-2 or worse defect and record at that bearing; the envelope top line should sit at the defect order (3.58× outer, 5.42× inner, 4.71× ball, 0.40× cage).

6. **Gear-mesh sidebands.** Fit the gearbox, crack a pinion tooth at 40 % or more, and record at the gearbox with F max 5 000 Hz.

7. **Sweep through resonance.** With some unbalance on the disc, sweep the speed through the pedestal natural frequency and record bearing 1 vertical near it.

8. **Diagnose three blind machines.** Load a blind machine, take a full route, and name the fault. Three correct diagnoses complete the task.

17 · Brake and Clutch Demonstrator

Folder: simulators/brake-clutch/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/brake-clutch/index.html

Topic: Friction devices · disc and drum brakes · plate clutch · torque capacity · energy and temperature rise · fade · clutch slip time

What it does: Disc brake (caliper pressure and bore, pad radii and arc), drum brake (leading-trailing shoes, self-energising) and plate clutch (spring force, plates). Uniform-wear or uniform-pressure friction radius, lining pressure against the allowable, torque capacity, brake factor. Stop or engagement transient integrated on the inertia: stop or slip time, energy absorbed, temperature rise per event on a thermal clock with cooling, friction fade above the lining's fade temperature, auto-repeat to ratchet the temperature, lock-up, continuous slip and engine stall for the clutch. Transient chart and temperature/friction history.

How to use it

1. Choose disc brake, drum brake or plate clutch, then set the actuation (pressure and bore, or spring force), the lining geometry and material, and the inertia or load.

2. Apply brake / Engage clutch runs one event: the transient chart shows speed, torque and lining temperature, the readouts give stop or slip time, energy and temperature rise. Auto-repeat runs events back to back so the lining heats and fades. The clock speeds the thermal history.

3. Bars on the right show lining temperature against the fade point and friction as a share of cold friction.

4. Record readings to compare distributions, pressures, disc against drum and cold against faded in the tasks. Every completed event is also logged.

Keys: Space run / pause · B apply / engage · A auto-repeat · R record · 1–3 mode · Esc close

Controls

Hydraulic pressure · Piston bore · Spring clamp force · Friction plates · Lining outer radius · Lining inner radius · Pad arc · Drum diameter · Shoe width · Lining material · Pressure distribution · Inertia being stopped · Speed before the stop · Engine speed · Engine torque during engagement · Engine inertia · Load inertia · Load torque · Pedal release time · Rotor mass (disc / drum / plates) · Cooling factor · Ambient · Auto-repeat interval · Clock

Action buttons: Start · Stop · Apply brake · Auto-repeat: off · Reset (cool, new lining) · Record reading

Readouts

Clamp / actuating force · Effective friction radius · Lining pressure (max) · Torque capacity · Friction coefficient · Energy per event · Temperature rise per event · Lining temperature · Last stop time · Events

Charts: Transient · speed, torque and lining temperature through the last stop or engagement · History · lining temperature, friction and speed over repeated events

Ready-made scenarios and presets

Single stop from 1 500 rpm · Repeated stops → fade · Small pad, high pressure · Drum brake, same actuation · Drum brake fade · Clutch engagement · Heavy load, slow release · Overloaded clutch, slips for ever · Dump the clutch → stall · Repeated engagements

Learning tasks (checked automatically)

1. **Make a stop.** Apply the disc brake from 1 500 rpm. Read the stop time, the energy absorbed and the temperature rise of the disc.

2. **Uniform wear against uniform pressure.** Record the same disc with both pressure distributions and compare the effective radius and torque. Which is the safe design assumption?

3. **Pressure to torque.** Record a brake at two hydraulic pressures with the same geometry: torque capacity scales with clamp force.

4. **Lining pressure limit.** Shrink the pad or raise the pressure until the lining pressure passes the allowable for the material.

5. **Fade.** Switch auto-repeat on and let the brake make repeated stops until friction falls below 80 % of cold. Watch the stop time grow.

6. **Drum self-energising.** Record a drum brake and a disc brake at the same pressure and piston bore (set the disc caliper to the drum bore). Compare the brake factor: the leading shoe multiplies the actuation.

7. **Drum sensitivity.** Record a drum brake cold and again when faded: because the leading shoe multiplies μ, its torque falls further than the disc brake does.

8. **Engage the clutch.** Release the clutch pedal: the engine droops, the load spins up, the plates lock. Read the slip time and the energy absorbed.

9. **Clutch that cannot lock.** Raise the load torque above the clutch capacity (or drop the spring force) and engage: the clutch slips continuously and burns.

18 · Fatigue and Stress Concentration Bench

Folder: simulators/fatigue-notch/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/fatigue-notch/index.html

Topic: Fatigue · stress concentration Kt and Kf · notch sensitivity · Marin factors · S-N curve · Goodman diagram · Miner damage · Paris crack growth

What it does: Stepped shaft (bending or axial) with a shoulder fillet, or a plate with a hole, in five materials with surface finish and reliability. Kt from chart fits, Peterson notch sensitivity, Kf, endurance limit with Marin factors, notch stresses with mean-stress relaxation at yield, Goodman/Gerber/Soderberg/ASME criteria, equivalent reversed stress and S-N life, cycles on a clock with Miner damage, Paris-law crack growth to the critical crack and fracture. S-N chart with smooth and derated curves and the cycle clock, Goodman diagram with all criteria, crack-growth chart.

How to use it

1. Choose a stepped shaft (bending or axial) or a plate with a hole, then the material, dimensions, fillet radius or hole size, finish and reliability.

2. Load it: amplitude and mean, the criterion, the frequency and the cycle clock. The readouts give Kt, q, Kf, Se, the notch stresses, factors of safety and the predicted life.

3. Start cycling: the damage bar fills, the crack initiates at the notch (drawn magnified), grows along the Paris curve and the specimen fractures. New specimen resets the clock.

4. Charts: the S-N diagram with the smooth-specimen and derated curves, the Goodman diagram with all criteria, and the crack-growth curve. Recorded readings appear as grey dots.

Keys: Space cycle / pause · N new specimen · R record · 1–3 geometry · Esc close

Controls

Material · Small diameter d · Shoulder ratio D/d · Fillet radius r · Plate width w · Plate thickness t · Hole diameter · Surface finish · Reliability · Moment amplitude Ma · Mean moment Mm · Mean-stress criterion · Load frequency · Cycle clock

Action buttons: Start cycling · Pause · New specimen · Record reading

Readouts

Stress concentration Kt · Fatigue concentration Kf · Endurance limit Se · Stress at the notch · Factor of safety · Predicted life · Cycles run · Crack length · Cycle rate · Specimens

Charts: S-N diagram · smooth specimen, this part, and where the load sits · Goodman diagram · mean stress against alternating stress at the notch · Crack growth · Paris law from initiation to the critical crack

Ready-made scenarios and presets

Sharp fillet, 1045 shaft · Fillet radius 4 mm · Ground finish · Add mean stress · Same shaft in 4340 · Infinite-life design · Aluminium, no endurance limit · Plate with a hole

Learning tasks (checked automatically)

1. **Run one to fracture.** Start cycling the baseline notched shaft and watch the damage bar, the crack initiate at the fillet and the specimen break. Compare the cycles run with the S-N prediction.

2. **Fillet radius buys life.** Record the same shaft and load with a fillet radius at least four times larger. The life should improve by more than three times, or go infinite.

3. **Surface finish.** Record the same part hot-rolled or as-forged and ground. Read what the surface factor ka does to the endurance limit and the life.

4. **Mean stress hurts.** Record the same amplitude at zero mean and at a mean load that gives at least 100 MPa mean stress at the notch. Watch the point move right on the Goodman diagram.

5. **Design for infinite life, efficiently.** Find a load or geometry that gives infinite life with the equivalent reversed stress at least 85 % of Se, and record it.

6. **Notch sensitivity.** Record the same geometry in two materials whose notch sensitivity q differs by 0.1 or more. The sharper, stronger steel feels the notch more fully.

7. **Watch the crack.** Let a specimen run until the crack passes 1 mm, and note how little of the life is left once a crack is visible.

8. **No endurance limit.** Record an aluminium part with a finite life above 10^7 cycles: the S-N curve never goes flat, so there is no stress that is safe for ever.

9. **Hole in a plate.** Record a plate whose hole gives Kt above 2.7 (a small hole) and one with Kt below 2.3 (a large hole) — the net section changes too.

19 · Bolted Joint and Torque Bench

Folder: simulators/bolted-joint/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/bolted-joint/index.html

Topic: Bolted joints · bolt and member stiffness · joint diagram · torque-tension and its scatter · embedding · separation · proof load · bolt fatigue

What it does: One bolt (M6–M24, 8.8/10.9/12.9) clamping two plates of steel, cast iron or aluminium with an optional gasket. Frustum member stiffness, bolt stiffness, C; preload from torque with the torque coefficient and scatter of the lubrication state, or by turn-of-nut or stretch; random tightenings on the torque-tension chart; embedding loss on a clock; static or cyclic external load with a load introduction factor; joint diagram with load partition; separation load, proof-load factor with bolt yield and preload loss, Goodman bolt fatigue factor. Load traces.

How to use it

1. Build the joint: bolt size and class, grip, member material, gasket and surface condition. The readouts show stiffnesses and C at once.

2. Tighten: choose the method and lubrication, set the torque (or the target preload) and press Tighten. Each tightening draws a random K from the scatter band; Tighten 20 bolts fills the torque–tension chart with dots.

3. Load it: set the maximum and minimum load and switch the cyclic load on. Watch the joint diagram partition the load, the clamp fall, the separation and proof margins, and the bolt fatigue factor.

4. Embedding sags the preload in the first minute after tightening; speed it up with the clock. Record readings to compare joints in the tasks.

Keys: Space run / pause · T tighten · M tighten 20 · L loosen · C cyclic load · R record · Esc close

Controls

Bolt size · Property class · Grip length (two equal plates) · Member material · Gasket · Surface condition (embedding) · Method · Wrench torque · Target preload · Thread lubrication (torque coefficient K) · Maximum load · Minimum load (cyclic) · Load frequency · Load introduction factor n · Embedding clock

Action buttons: Running · Pause · Tighten · Tighten 20 bolts · Loosen · Cyclic load: off · Record reading · Clear samples

Readouts

Bolt · Stiffness ratio C · Preload from this setting · Preload now · Bolt load · Clamp on the members · Separation load · Proof-load factor · Bolt fatigue factor · Recommended torque

Charts: Joint diagram · bolt stretch against member compression, preload and load partition · Torque against tension · scatter of tightenings · Loads · bolt, clamp, external and preload over time

Ready-made scenarios and presets

M12 8.8, 90 N·m, zinc plated · 20 bolts, dry threads · Same torque, MoS_2 → yield · Rubber gasket · Painted surfaces, embedding · Overload → separation · Cyclic load on a soft joint · 20 bolts by stretch

Learning tasks (checked automatically)

1. **Tighten a bolt.** Tighten the M12 with the torque wrench and read the preload, the bolt stress as a share of proof and the load partition on the joint diagram.

2. **Torque is not tension.** Set the threads to dry and tighten 20 bolts at the same torque. Read the spread of preload: this is why torque control alone is ±35 %.

3. **Lubrication changes everything.** Record a tightening dry and one with MoS_2 at the same torque. The lubricated bolt sees far more preload — enough to yield if the torque was set for dry threads.

4. **Gasket stiffness.** Record the joint metal-to-metal and again with a rubber gasket. Compare C: the soft gasket hands most of the external load to the bolt.

5. **Embedding loss.** Tighten a joint with painted or rough surfaces and watch the preload sag by 5 % or more in the first minute as the surfaces bed in.

6. **Separate the joint.** Raise the external load past the separation load: the clamp goes to zero and the bolt carries the whole load, cycle after cycle.

7. **Yield the bolt.** Over-torque, or lubricate a bolt torqued for dry threads, until the bolt load passes the proof load. Note the preload that is lost.

8. **Fatigue of the bolt.** With a cyclic load, find a joint whose bolt fatigue factor is below 1 and record it; then fix it (stiffer joint, more preload, bigger bolt) to a factor of 1.5 or more and record that.

9. **A better method.** Switch to turn-of-nut or bolt stretch and tighten 20 bolts. Compare the spread with the torque wrench.

20 · Steam and Gas Turbine Cycle Board

Folder: simulators/turbine-cycles/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/turbine-cycles/index.html

Topic: Thermodynamic cycles · Rankine with superheat, reheat and condenser vacuum · Brayton with pressure ratio, turbine inlet temperature and regenerator · combined cycle · T-s diagrams

What it does: Rankine steam cycle (boiler pressure, superheat, condenser pressure, turbine and pump efficiencies, reheat) on a saturation table with an effective superheat heat capacity; Brayton gas turbine (pressure ratio, ambient, turbine inlet temperature, component efficiencies, regenerator) on the air standard; combined cycle through an HRSG. Flow board with state points, T-s diagram, parameter sweeps of efficiency and exhaust quality or net work, heat rate and mass flow for a plant output. Alarms for wet exhaust, turbine inlet temperature and useless regenerator.

How to use it

1. Choose the Rankine steam cycle, the Brayton gas turbine or the combined cycle. The flow board shows the state points with temperature, pressure, enthalpy and quality.

2. Move the sliders: boiler pressure, superheat, condenser pressure, reheat, component efficiencies, pressure ratio, turbine inlet temperature, regenerator. Everything updates at once, including the T-s diagram and the sweep chart.

3. Sweep chart: efficiency against the main design variable with the present setting marked, so you can see where you sit on the curve.

4. Record readings to compare cycles in the tasks.

Keys: Space flow / pause · R record · 1–3 cycle · Esc close

Controls

Boiler / HRSG pressure · Turbine inlet temperature · Condenser pressure · Steam turbine isentropic efficiency · Feed pump efficiency · Reheat · Reheat pressure · Pressure ratio · Ambient (compressor inlet) · Compressor isentropic efficiency · Gas turbine isentropic efficiency · Regenerator effectiveness · Plant output

Action buttons: Flowing · Pause flow · Record reading

Readouts

Thermal efficiency · Net work · Heat added · Turbine work · Work to the pump · Turbine exhaust quality · Boiler saturation temperature · Flow for the plant output · Heat rate

Charts: T-s diagram · steam dome and the Rankine cycle · Sweep · efficiency and exhaust quality against boiler pressure

Ready-made scenarios and presets

Rankine 8 MPa / 500 °C / 10 kPa · Boiler to 16 MPa · Condenser to 5 kPa · 16 MPa with reheat · Turbine efficiency 0.75 · Gas turbine rp 10, 1 200 °C · rp 6 with regenerator · rp 30 · Combined cycle

Learning tasks (checked automatically)

1. **Baseline Rankine cycle.** Record the 8 MPa / 500 °C / 10 kPa cycle. Compare its efficiency with the Carnot limit between the same temperatures and note where the loss is.

2. **Raise the boiler pressure.** Record two Rankine cycles whose boiler pressures differ by 4 MPa or more with the same superheat temperature: efficiency rises, the exhaust gets wetter.

3. **Superheat dries the exhaust.** Record the same pressure with turbine inlet temperatures 100 K apart: the exhaust quality improves and so does efficiency.

4. **Condenser vacuum.** Record condenser pressures 40 kPa apart: a better vacuum lowers the heat-rejection temperature and lifts efficiency, at the price of a wetter exhaust and a bigger condenser.

5. **Reheat cures wetness.** At 12 MPa or more, switch reheat on and record a cycle whose exhaust quality is at least 0.90.

6. **Turbine efficiency.** Record the same cycle with turbine isentropic efficiencies 0.10 apart. Irreversibility dries the exhaust but costs work.

7. **Pressure ratio optimum.** Brayton: find the pressure ratio for maximum net work (within 20 % of the optimum) and record it; then note that efficiency keeps rising beyond it.

8. **Regenerator.** Brayton at a low pressure ratio (6): record without and with a regenerator of effectiveness 0.6 or more. Where does the recovered heat show on the T-s diagram?

9. **Combined cycle above 50 %.** Combine the two: choose the pressure ratio, turbine inlet temperature and steam conditions that take the combined efficiency to 50 % or more, and record it.

21 · Control Valve and PID Loop Bench

Folder: simulators/control-valve-pid/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/control-valve-pid/index.html

Topic: Process control · control valve characteristics and authority · actuator lag and stiction · PID with anti-windup · step-test identification · tuning rules

What it does: Flow, level or temperature loop closed by an ISA PID in manual or auto. Control valve with linear, equal-percentage or quick-opening characteristic, Kv, system resistance (installed characteristic and authority), actuator lag, stiction and a positioner. Dead time, process lag, noise and disturbances. Step tests identify FOPDT or integrating models and suggest lambda, SIMC or Ziegler–Nichols tuning; overshoot, settling time, IAE and oscillation period are measured. Trends, installed-characteristic chart and OP-against-stem trace.

How to use it

1. Manual first: the controller starts in manual. Move the manual output and watch the valve stem, the flow and the PV respond. A step of 10 % and a minute of patience gives an identified process model and a suggested tuning.

2. Auto: switch the mode, set Kc, Ti and Td, and step the setpoint with the +10 / −10 buttons. The readouts give overshoot, settling time and IAE for the last step; oscillation is detected and its period shown.

3. Valve: change the characteristic, the system resistance (authority) and stiction, and watch the installed characteristic chart and the OP-against-stem trace.

4. Disturbances: move the pump pressure or the load and watch the loop recover. Record readings to compare tunings and valves in the tasks.

Keys: Space run / pause · A auto / manual · + / − setpoint ±10 · R record · Esc close

Controls

Inherent characteristic · Valve Kv (full open) · Pipe / system resistance (lowers authority) · Actuator time constant · Stiction (deadband) · Positioner · Setpoint · Manual output · Gain Kc · Integral time Ti (0 = off) · Derivative time Td · Anti-windup · Controller action · Dead time · Process time constant · Measurement noise · Pump pressure (disturbance) · Downstream pressure · Load (tank outflow / product flow) · Clock

Action buttons: Running · Pause · Mode: MANUAL · SP +10 · SP −10 · Apply suggested tuning · Record reading · Clear trends

Readouts

Process variable · Error · Controller output · Flow · Valve Δp and authority · Last step · overshoot · Oscillation · Identified process · Suggested Kc / Ti / Td

Charts: Trends · PV, SP, OP and stem · Valve characteristic · inherent (grey) and installed (blue) · Controller output against stem · lag and stiction

Ready-made scenarios and presets

Manual step test · P-only: offset · PI, well tuned · Gain too high · Windup (SP 100 %) · Stiction 2 % · Linear valve, low authority · Temperature loop, dead time · Level loop (integrating)

Learning tasks (checked automatically)

1. **Open-loop step test.** In manual, step the output by 10 % and let the response settle. Record the identified process gain, time constant and dead time.

2. **Proportional-only offset.** Switch to auto with integral off (Ti = 0). The loop settles short of setpoint: read the offset and note how raising Kc shrinks but never removes it.

3. **Integral removes the offset.** Give the controller integral action (Ti 5–20 s) and hold within 0.5 % of setpoint for 8 s.

4. **Too much gain.** Raise Kc until the loop oscillates steadily. That gain is Ku and the period is Pu — the Ziegler–Nichols closed-loop data.

5. **Apply a tuning.** Apply the suggested tuning (from the step test or from Ku and Pu) and step the setpoint 10 %: overshoot under 25 % and a measured settling time.

6. **Integral windup.** With anti-windup off and the pump at 5 bar, set a setpoint the valve cannot reach so the output saturates for 20 s or more, then set the setpoint to 80 %: the output stays saturated while the integral unwinds and the PV hangs above setpoint. Switch anti-windup on and repeat.

7. **Stiction limit cycle.** Give the valve 2 % stiction in auto with integral action and watch the stick-slip limit cycle; then enable the positioner.

8. **Characteristic and authority.** Record the loop with a linear valve and with an equal-percentage valve at an authority below 0.3. Compare the installed gain at the operating point.

9. **Dead time.** Set 5 s or more of dead time and find a tuning that still settles with less than 25 % overshoot. Record it.

22 · Refrigeration Cycle Trainer

Folder: simulators/refrigeration-cycle/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/refrigeration-cycle/index.html

Topic: Vapour-compression refrigeration · p-h diagram · COP · superheat and subcooling · compressor volumetric efficiency · cold-room dynamics and thermostat cycling

What it does: R134a cycle with evaporating and condensing temperatures, suction superheat, liquid subcooling, compressor isentropic efficiency, swept volume, clearance and charge: capacity, power, COP against Carnot, mass flow, discharge temperature, flash gas, volumetric efficiency on a p-h diagram with a capacity/COP sweep. Cold-room mode: a lumped room pulled down from ambient with heat leak and product load, evaporating temperature following the room, condensing temperature following the ambient and condenser fouling, thermostat cycling on a clock.

How to use it

1. Cycle explorer: set evaporating and condensing temperatures, superheat, subcooling and compressor data. The circuit, the p-h diagram and the readouts update at once.

2. Cold room: the room starts at ambient; the compressor pulls it down while the evaporating temperature follows the room and the condensing temperature follows the ambient. The thermostat cycles the compressor. The clock compresses hours into minutes.

3. Sweep chart: capacity and COP of this compressor against evaporating temperature at two condensing temperatures, with the present cycle marked.

4. Record readings to compare cycles in the tasks.

Keys: Space run / pause · R record · 1 explorer · 2 cold room · Esc close

Controls

Evaporating temperature · Condensing temperature · Suction superheat · Liquid subcooling · Compressor isentropic efficiency · Swept volume · Clearance volume · Refrigerant charge · Ambient at the condenser · Room thermostat setting · Product / door load · Room heat leak UA · Room thermal mass · Condenser fouling (extra approach) · Evaporator approach · Condenser approach · Clock

Action buttons: Running · Pause · Warm room to ambient · Record reading

Readouts

Cooling capacity · Compressor power · COP · Evaporating · Condensing · Discharge temperature · Superheat / subcooling · Flash gas after the valve · Volumetric efficiency · Cold room

Charts: p-h diagram · R134a with the cycle · Capacity and COP against evaporating temperature · Cold room · temperature, capacity and power against time

Ready-made scenarios and presets

−10 / 40 °C, 5 K superheat · Condensing 55 °C · Evaporating −30 °C · No superheat · Pressure ratio too high · Low charge 60 % · Cold room pull-down · Hot day, fouled condenser

Learning tasks (checked automatically)

1. **Read the cycle.** Record the −10 / 40 °C cycle: capacity, power, COP, and how far it sits below the Carnot COP.

2. **Condensing temperature.** Record the same evaporating temperature with condensing temperatures 15 K apart: a hot condenser costs COP and capacity.

3. **Evaporating temperature.** Record evaporating temperatures 20 K apart at the same condensing temperature: capacity collapses at low temperature because the suction gas is thin.

4. **No superheat.** Set the superheat to zero (or overcharge) and see the suction state land inside the dome: liquid slugging.

5. **Discharge too hot.** Push the pressure ratio up (deep freeze evaporating, hot condenser) until the discharge passes 120 °C.

6. **Subcooling adds capacity.** Record 0 K and 10 K of subcooling at the same temperatures: the extra liquid enthalpy goes straight into capacity at no compressor cost.

7. **Pull the room down.** Cold-room mode: start at ambient and pull the room to its set point. Read how the capacity falls as the evaporating temperature drops.

8. **Thermostat cycling.** Let the cold room cycle on the thermostat at least three times and read the run fraction from the shaded trace.

9. **Fouled condenser.** In the cold room add 8 K or more of condenser fouling and record: condensing temperature, power and discharge temperature rise, capacity falls.

23 · Rotor Balancing and Critical Speed Rig

Folder: simulators/rotor-balancing/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/rotor-balancing/index.html

Topic: Rotor dynamics · Jeffcott rotor · critical speed and phase · damping · run-up · single-plane influence-coefficient balancing · ISO balance grade

What it does: Disc on a flexible shaft between two bearings with adjustable mass, shaft diameter, span, disc position, support stiffness and damping. Unbalance response amplitude and phase, run-up sweep through the critical with resonance build-up, whirl orbit with heavy and high spot, vibration trip. Balancing job with a hidden unbalance: original and trial-mass readings on a polar plot, computed correction, residual and ISO balance grade at service speed. Bode plot, polar plot and amplitude trend.

How to use it

1. Rotor: set disc mass, shaft diameter, span, disc position, bearing stiffness and damping. The critical speed updates at once.

2. Run-up sweep drives the speed up through the critical and back; watch the Bode plot fill, the orbit grow and the phase swing. The trip limit stops the drive if the vibration is too high.

3. Balancing job: at a speed well below the critical take the original reading, fit the trial mass, take the trial reading, use the computed correction (or compute your own from the polar plot), fit it and check the residual.

4. Record readings to compare damping, supports and speeds in the tasks.

Keys: Space run / pause · S sweep · O original reading · T trial reading · R record · Esc close

Controls

Disc mass · Shaft diameter · Bearing span · Disc position · Bearing / support stiffness · Damping ratio ζ · Speed setpoint · Ramp rate · Vibration trip limit · Trial mass · Trial mass angle · Trial mass radius · Correction mass · Correction angle · Correction radius

Action buttons: Running · Pause · Run-up sweep · Reset trip · Take original reading · Fit trial mass · Take trial reading · Use computed correction · Fit correction mass · New unknown unbalance · Record reading

Readouts

Speed · Vibration · Stiffness · Residual unbalance · Balance grade at service speed · Original reading O · Trial reading O + T · Computed correction

Charts: Bode plot · amplitude and phase against speed · Polar plot · influence-coefficient balancing · Trend · amplitude and speed

Ready-made scenarios and presets

Run-up through the critical · Fast ramp, high trip limit · Damping 0.2 · Soft supports · Balancing job at 1 200 rpm · Run above the critical

Learning tasks (checked automatically)

1. **Find the critical speed.** Run a sweep and watch the amplitude peak and the phase pass through 90°. Compare the peak speed with the calculated critical.

2. **Phase across the critical.** Record readings below (ratio < 0.7) and above (ratio > 1.4) the critical: the phase lag goes from near 0° to near 180° and the disc whirls about its mass centre.

3. **Damping and the peak.** Record the critical-speed amplitude at damping ratios 0.05 and 0.15 or more: the peak scales with 1/2ζ.

4. **Vibration trip.** Drive the rotor through the critical with the raw unbalance and trip it on the vibration limit; then raise the ramp rate and see whether you can get through.

5. **Original reading.** At the balancing speed (well below the critical) take the original vibration reading: amplitude and phase.

6. **Trial mass.** Fit a trial mass and take the trial reading at the same speed. The polar plot shows the effect vector T.

7. **Balance it.** Fit the computed correction mass at its angle and get the residual vibration below 20 % of the original.

8. **Balance grade.** Bring the residual unbalance to grade G2.5 or better at the service speed and record it.

9. **Soft supports.** Record the critical speed on stiff bearings and again with the bearing stiffness reduced to 10 MN/m or less: the supports lower the critical.

24 · Shaft Alignment Bench

Folder: simulators/shaft-alignment/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/shaft-alignment/index.html

Topic: Shaft alignment · offset and angularity · rim-and-face and reverse-dial readings · bracket sag · shims and moves · thermal growth · soft foot · tolerances by speed · vibration from misalignment

What it does: Pump and motor with hidden as-found errors in both planes. Rim, face and reverse-dial readings with bracket sag, a laser display, and shim or jacking corrections at the front and rear feet; thermal-growth targets, a soft-foot check and fix, speed-dependent tolerances. Running the set gives coupling reaction, 1× and 2× vibration in ISO zones, bearing life and heating. Alignment graph of the centrelines, vibration bars and trend.

How to use it

1. Read the rim and face dials (or the reverse-dial and laser readouts) in the vertical and horizontal planes. Mind the bracket sag.

2. Correct: work out the shims at the front and rear feet from the foot distances, then apply them with the sliders; switch to the plan view and jack the motor sideways. Show solution reveals the answer, Apply does it for you.

3. Before you start run the soft-foot check; if a foot shows a gap, shim it first. If the motor grows when hot, align to the hot target.

4. Start the set to see what the misalignment does: 1× and 2× vibration, coupling reaction, bearing temperature. Record readings for the tasks.

Keys: Space start / stop · V vertical · H horizontal · R record · Esc close

Controls

Coupling to front feet · Coupling to rear feet · Face indicator diameter · Reverse-dial span · Running speed · Shims under the front feet · Shims under the rear feet · Sideways move at the front feet · Sideways move at the rear feet · Shim under the soft foot · Indicator bracket sag · Sag compensation · Motor thermal growth (rises when hot) · Alignment target · Running clock

Action buttons: Start the set · Stop · Soft-foot check · Show solution · Apply the computed corrections · New random job · Record reading

Readouts

Vertical offset / angularity · Horizontal offset / angularity · Tolerance at this speed · Rim readings V / H · Face readings V / H · Reverse-dial rims · Offset · angle from the dials · Laser display (true) · Running vibration 2× · Soft-foot check

Charts: Alignment graph · shaft centrelines and feet · Vibration at the motor bearing · 1×, 2×, 3× · Trend · vibration and bearing temperature

Ready-made scenarios and presets

As-found job at 1 500 rpm · Random as-found job · Hot motor, align for growth · Soft foot hidden in the job · 3 600 rpm tolerance · Run it as found

Learning tasks (checked automatically)

1. **Read the as-found condition.** Record the as-found readings before touching anything: rim and face in both planes, and the offset and angularity they imply.

2. **Shim the vertical plane.** Work out the shim change at the front and rear feet from the readings and the foot distances, apply it and bring the vertical plane within tolerance.

3. **Move the horizontal plane.** Jack the motor sideways at each foot to bring the horizontal offset and angularity within tolerance.

4. **Aligned in both planes.** Both planes within the tolerance for the running speed. Record it.

5. **Bracket sag.** With bracket sag not compensated, record the offset the dials imply; then switch compensation on and record again. The difference is exactly the sag.

6. **Thermal growth.** Set a motor thermal growth of 0.05 mm or more, align for the hot condition (cold target below zero) and get the hot alignment within tolerance.

7. **Soft foot.** Run the soft-foot check, find the foot with a gap, shim it and confirm the gap is closed before aligning.

8. **Run it misaligned.** Start the set with the misalignment three times the tolerance or more and watch the 2× vibration climb into zone C and the bearing heat.

9. **Tighter at speed.** Set 3 000 rpm or more and align to the tighter tolerance that applies.

25 · Gearbox Tooth Loads and Efficiency Bench

Folder: simulators/gearbox-teeth/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/gearbox-teeth/index.html

Topic: Gear design · tooth forces · Lewis bending and Hertz contact stress · contact ratio · dynamic factor · mesh, churning, bearing and seal losses · oil temperature · pitting and scuffing

What it does: Spur or helical pair from module, teeth, face width, helix angle and quality in four materials. Tangential, radial and axial forces and bearing loads; bending and contact stresses with safety factors against the allowables; contact and overlap ratios; efficiency from mesh friction, churning, bearing and seal losses at any load and speed; oil temperature on a clock with viscosity feedback, film ratio, pitting damage and scuffing. Stress-against-power and efficiency-against-load charts, loss breakdown and oil trend.

How to use it

1. Geometry: module, teeth, face width, helix and quality. The drawing, forces and stresses update at once; the stress chart shows both stresses against power with their allowables.

2. Drive: speed, power and service factor. The efficiency chart shows where you sit on the load curve and how the losses split.

3. Run the clock to heat the oil: viscosity falls, friction and film ratio change, and overloads use up pitting life or scuff the teeth. New gears and oil resets damage and temperature.

4. Record readings to compare module, face width, helix, load and speed in the tasks.

Keys: Space run / stop · N new gears · R record · Esc close

Controls

Module · Pinion teeth · Gear teeth · Face width · Helix angle · Manufacturing quality · Material (both gears) · Pinion speed · Transmitted power · Overload / service factor Ko · Oil grade · Oil level · Ambient · Clock

Action buttons: Running · Stop · New gears and oil · Record reading

Readouts

Ratio and output · Pitch line velocity · Tooth forces · Bending stress · Contact stress · Contact ratio · Efficiency · Oil temperature · Bearing loads · Pitting damage

Charts: Tooth stresses against transmitted power · Efficiency against load · loss breakdown · Trend · oil temperature and efficiency

Ready-made scenarios and presets

Spur m4 20:60, 30 kW · Same pair, helical 20° · Module 6 · Part load 6 kW · 3 000 rpm churning · Overload 75 kW → pitting · Thin hot oil → scuffing · Cast iron gears

Learning tasks (checked automatically)

1. **Tooth forces.** Record the baseline pair (m 4, 20:60, 50 mm face, 30 kW at 1 450 rpm): tangential, radial and axial force and the bearing loads.

2. **Module against bending.** Record the same power and teeth with a module 2 mm larger: the tangential force falls with the larger pitch diameter and the tooth is thicker — bending stress drops sharply.

3. **Face width.** Record face widths 20 mm apart at the same load: both stresses fall, contact stress with the square root.

4. **Helix angle and thrust.** Set a helix angle of 15° or more: the mesh is smoother (overlap ratio) and the stresses fall, but an axial thrust appears that the bearings must carry.

5. **Which stress governs?.** Find a load where the contact safety factor is below 1.2 while bending is still above 1.5 — pitting governs most steel gears. Record it.

6. **Efficiency at part load.** Record the pair at full power and at 20 % power: the fixed churning and seal losses make part-load efficiency worse.

7. **Churning at speed.** Run at 2 500 rpm or more and record: churning grows faster than the transmitted power. Compare a thinner oil.

8. **Overload to pitting.** Overload the pair until the contact stress passes the allowable, run the clock and watch pitting damage accumulate to failure.

9. **Scuffing.** Combine high load, high speed and thin hot oil (VG 68, ambient 45 °C) until the film ratio collapses and the teeth scuff.

26 · API 650 Tank Design Demonstrator

Folder: simulators/api650-tank/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/api650-tank/index.html

Topic: Storage tank design · API 650 one-foot method · allowable stresses and materials · corrosion allowance · plate schedule · bottom and annular plates · roofs · wind girders · overturning and anchorage · steel mass against diameter · hydrotest

What it does: Welded atmospheric storage tank sized course by course with the one-foot method in six materials with corrosion allowance, standard plate rounding, minimum thickness, no-thinner-than-above and 45 mm rules; manual thickness overrides per course with the checks failing visibly. Bottom and annular plates from the first-course stress, supported cone, self-supporting cone or dome roof plate with the 13 mm feasibility limit and junction ring force, top and intermediate wind girders by the transformed-shell method, wind overturning with the unanchored-tank criteria. Steel masses, a constant-capacity steel-against-diameter sweep, and fill or hydrotest on a clock with the live hoop stress in every course.

How to use it

1. Size the tank: diameter, number and height of courses, design liquid level, product gravity, corrosion allowance and material. The plate schedule, steel mass and every check update at once; the elevation shows each course with its ordered thickness.

2. Read the charts: the schedule chart shows required against ordered per course and which case governs; the utilisation chart shows product and test stresses against the allowables; the sweep shows the steel needed at every diameter for this capacity.

3. Override: click a course, then Thicker / Thinner (or the arrow keys) to set the plate by hand and see the checks fail or pass. Auto returns that course to the calculated plate.

4. Roof and wind: choose a roof and slope, set the wind. Watch for intermediate girders, the anchorage verdict and roof feasibility.

5. Fill and test: Fill to H with product, or Hydrotest with water, on a fast clock. The live stress in every course is drawn on the shell and on the utilisation chart.

Keys: Space run / pause · F fill · W hydrotest · E drain · ↑ / ↓ thicker / thinner on the selected course · A auto · R record · Esc close

Controls

Diameter D · Shell courses · Course height · Design liquid level H · Product specific gravity G · Corrosion allowance · Shell material (Table 5-2 allowables) · Roof · Roof slope · Dome radius · Roof live load · Ground snow load · Design wind speed (3 s gust) · Fluid · Fill rate · Clock

Action buttons: Running · Pause · Fill to H · Hydrotest · Drain · Thicker ▲ · Thinner ▼ · Auto · Reset schedule · Record reading

Readouts

Working capacity · Bottom course · Plate schedule · Steel · Bottom plates · Roof plate · Intermediate wind girders · Anchorage · Live stress, bottom course · Selected course

Charts: Shell plate schedule · required against ordered, course by course · Stress utilisation · design and test cases against the allowables, live fill marker · Steel against diameter at constant capacity

Ready-made scenarios and presets

30 m × 14.4 m, A283 C, cone roof · Same tank in A516 Gr 485 · 3 mm corrosion allowance · Heavy product G 1.25 · Hydrotest on a fast clock · Tall open-top 12 m × 24 m · 45 m open top in 230 km/h · 6 m × 14.4 m empty in wind · Self-supporting cone on 30 m · 30 m dome roof · 70 m tank in A537 · 10 m water tank

Learning tasks (checked automatically)

1. **Read the baseline.** Record the 30 m tank as it opens. Read the plate schedule from the bottom course up, the total steel and the kg per m^3 of capacity.

2. **The one-foot rule.** Click the bottom course and the second course. Compare the required thickness with the height of liquid above each course: the design point sits 0.3 m above the bottom of the course, where the course above still helps.

3. **Who governs?.** With the light product (G 0.85) the hydrotest governs the bottom course. Raise G until the product case governs and record both readings.

4. **Buy a better plate.** Record the same tank in A283M Gr C and again in A516M Gr 485 or A537M. The shell steel falls by 10 % or more — and the annular plate appears.

5. **Corrosion allowance.** Record the tank with no corrosion allowance and again with 3 mm. Note that the allowance adds to the product case only, never to the hydrotest case.

6. **Squat or tall?.** Use the sweep chart: at this capacity, find the diameter that needs the least steel. Record two tanks of the same capacity (within 3 %) at least 6 m apart in diameter.

7. **Stiffen the shell.** Make the tank tall and thin, or raise the wind speed, until an intermediate wind girder is required. Record it and read where it sits on the transformed shell.

8. **Blow it over.** Record a small, tall, empty tank in a strong wind that needs anchors; then change the design (lower, wider, heavier roof) until it stands unanchored and record that.

9. **Break a course and fix it.** Select a course and make it thinner than required, or thinner than the course above it. See the check fail. Then bring it back to a passing schedule.

10. **Hydrotest.** Press Hydrotest: the tank fills with water to the design level. Watch the live stress in the bottom course approach the test allowable St and record the reading at full.

11. **Roof choice.** Put a self-supporting cone on the 30 m tank at 9.5°: the plate needs more than 13 mm. Record it, then record a feasible roof (supported cone or dome) on the same diameter.

12. **Beyond the one-foot method.** Take the diameter past 61 m and read the warning: the one-foot method is no longer permitted and the variable-design-point method (or Annex A rules for small tanks) takes over.

27 · API 620 Tank Design Demonstrator

Folder: simulators/api620-tank/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/api620-tank/index.html

Topic: Low-pressure storage tanks · API 620 membrane design · unit forces T1 and T2 · dome, torispherical and cone roofs · knuckle compression · compression-ring region · buckling allowable · uplift and anchor bolts · Annex R and Q cold service · vapour space, vent and hydro-pneumatic test

What it does: Large welded tank with a gas space up to 103 kPa gauge, designed by membrane theory: meridional and hoop unit forces at every level of a dome, torispherical or cone roof and the sidewall for the design-pressure-with-liquid case and the vacuum-with-live-load case; tension thickness T/(S E) + CA and compression thickness from the t/R buckling allowable; the knuckle in hoop compression; the compression-ring region with participating widths and the area to add; uplift and anchor bolts at design and 1.25 × test pressure; scope limits (15 psig, 121 °C) and Annex R / Q temperature ranges with minimum design metal temperature per material; steel mass and a pressure sweep. Fill, pump out, pressurise, hydro-pneumatic test and blocked-vent operation on a clock with an isothermal vapour-space model, PV valve, live utilisation of every element and failure at 1.5 × allowable.

How to use it

1. Define the tank: diameter, courses, liquid level, product (which sets gravity, temperature and a typical gas pressure), then pressure, vacuum, material, radiography and corrosion allowance. Every plate, the ring, uplift and the checks update at once.

2. Read the forces: the first chart follows T1 and T2 from the apex of the roof down to the bottom for the pressure case (solid) and the vacuum case (dashed). Shaded values are compression.

3. Roof choice: dome, torispherical or cone. The knuckle or the compression ring appears on the drawing; set the ring section with its slider until the region passes.

4. Override: click a course, the roof or the knuckle, then Thicker / Thinner or the arrow keys. Auto returns to the calculated plate.

5. Operate: Fill, Pump out, Pressurise, the hydro-pneumatic test (water to 1.25 × G × H, then air to 1.25 × Pg, utilisation shown against 1.25 × the allowable), or block the vent and watch the vapour-space pressure on the gauge. Elements turn hot as they pass their allowable and the tank fails at 1.5 ×.

Keys: Space run / pause · F fill · E pump out · P pressurise · W test · V block / open vent · ↑ / ↓ thicker / thinner · A auto · R record · Esc close

Controls

Diameter D · Sidewall courses · Course height · Design liquid level H · Product (sets G, temperature, pressure) · Specific gravity G · Design metal temperature · Design gas pressure Pg · Design vacuum Pe · Material and allowable Sts · Joint efficiency E · Corrosion allowance · Roof · Crown radius Rc · Knuckle radius rk · Cone slope · Roof live load · Snow load · Compression ring section fitted · Anchor bolt · Bolts fitted · Fluid · Pump rate · Clock

Action buttons: Running · Pause · Fill to H · Pump out · Pressurise to Pg · Hydro-pneumatic test · Vent: normal · Reset process · Thicker ▲ · Thinner ▼ · Auto · Reset schedule · Record reading

Readouts

Working capacity · Bottom course · Sidewall schedule · Roof plate · Knuckle plate · Compression ring area to add · Uplift and anchors · Steel · Scope and material · Live utilisation, worst element · Selected element

Charts: Unit forces T1 and T2 along the meridian · design pressure with liquid, and the vacuum case · Plate schedule · roof, knuckle and sidewall courses, required against ordered · Steel and anchor bolts against design gas pressure

Ready-made scenarios and presets

30 m ammonia, 10 kPa, dome · Pentane at 50 kPa · Torispherical roof · 15° cone roof · 35 kPa blanket, too few bolts · 6 kPa design vacuum · 40 m LNG, 9 % Ni · 110 kPa: past 15 psig · Hydro-pneumatic test · Blocked vent, pump out · Blocked vent, fill · 12 m naphtha tank

Learning tasks (checked automatically)

1. **Read the baseline.** Record the 30 m refrigerated ammonia tank as it opens: plate schedule, roof plate, compression ring, uplift and anchors, total steel.

2. **T1 and T2 along the meridian.** Click the bottom course, the roof and the ring. On the unit-force chart follow T1 and T2 from the apex, across the junction and down the sidewall; note where the hoop force is set by gas pressure and where the liquid head takes over.

3. **Pressure against head.** Record the tank at 10 kPa and again at 50 kPa. Compare the bottom-course hoop force T2: at 10 kPa the liquid head dominates, at 50 kPa the gas pressure does, and every course above the liquid grows.

4. **The knuckle in compression.** Switch to the torispherical roof. Read T2 at the junction: negative — the knuckle is in hoop compression under internal pressure, and its plate is sized by buckling. Shrink the knuckle radius and watch it grow.

5. **Size the compression ring.** On the dome roof, set the ring-area slider below the area needed, see the region fail, then bring it up to pass. Record the passing design.

6. **Why not a cone?.** Put a 15° cone on the same tank at the same pressure and record it. Its hoop force is R/sin θ times the pressure: the roof plate and the ring both grow, and the vacuum case is worse still.

7. **Hold it down.** Raise the pressure until the empty tank lifts: uplift Pg × area exceeds the metal weight. Fit enough anchor bolts of a size that pass at design and test pressure, with spacing under 3 m, and record it.

8. **Vacuum design.** Raise the design vacuum to 5 kPa or more and record it. The top courses and the roof are now sized by compression, not tension: the buckling allowable depends on t/R.

9. **Cold service.** Choose LNG (−162 °C) with a carbon steel and read the material warning; then pick an Annex Q material and record a passing design.

10. **Fifteen psig.** Push the gas pressure past 103.4 kPa and read the scope warning: above 15 psig the tank is a pressure vessel, not an API 620 tank. Then bring it back inside.

11. **Hydro-pneumatic test.** Run the test: water to the test level (1.25 × G × H, so the head matches 1.25 × the product load), vent gagged, then air to 1.25 × Pg. Watch the utilisation of every element and record the reading at test pressure.

12. **Blocked vent.** Block the vent and pump out: the vapour space goes into vacuum within seconds and the sidewall buckles. Then block it and fill: pressure climbs past the setpoint until something yields. Record one failure.

28 · Gas Pipeline Route and Wall Thickness Designer

Folder: simulators/gas-pipeline-route/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/gas-pipeline-route/index.html

Topic: Gas transmission pipelines · ASME B31.8 · location class by building count · design factors and exceptions · Barlow design formula with F, E, T · standard walls and handling D/t · test pressure · class location change of an existing line · potential impact radius and HCAs

What it does: A 12 km onshore gas line through farms, a village and a town. A sliding 1-mile, 1/4-mile-wide unit counts buildings and sets Location Class 1 to 4 (with the 660 ft cluster rule); design factors from Tables 841.114A/B with road crossings, metering stations and concentrations of people; t = PD/(2SFET) rounded to standard walls with a handling D/t limit; test pressure by class; fracture-control threshold; PIR and HCA length. Build the line, let the town grow, and check the MAOP against Table 854.1(c) and 855.1; comply by derating or replacing pipe. Clause numbers follow ASME B31.8-2003.

How to use it

1. Read the route: houses, a village and a town lie along a 12 km line. Move over the map: the dashed box is the 1-mile, ¼-mile-wide class-location unit, and the readout counts the buildings in it.

2. Design: choose pipe size, grade, MAOP, temperature and seam. Every 10 m of route gets a location class, a design factor and a wall from the standard list. The strips under the map show class, F, wall and high consequence areas.

3. Change the route: pick a tool and click the map to add houses, an estate, flats, a school, a road crossing or a station. Erase removes the nearest item.

4. Build and operate: Build freezes the walls and records the test pressure. Then add buildings, or press +5 years, and see whether the MAOP is still allowed by Table 854.1(c). Reduce the MAOP or replace the pipe to comply.

5. Record readings to compare designs and to tick off the learning tasks.

Keys: B build · Y +5 years · R record · 1–8 tools · Esc close

Controls

Outside diameter · Line pipe grade (API 5L, SMYS) · MAOP · Design temperature · Longitudinal seam (Table 841.115A) · Class 1 design factor · Corrosion allowance · Handling limit D/t (company rule) · Class boundaries

Tools: Inspect unit · House · 10-house estate · Multistorey block · School (20+ people) · Road crossing · Metering station · Erase

Action buttons: Build pipeline · +5 years development · Reduce MAOP to comply · Replace non-compliant pipe · Back to design · Record reading

Readouts

Class-location unit at pointer · Route by location class · Wall thicknesses used · Line pipe steel · Hoop stress at MAOP · Test pressure (Table 841.322(f)) · Potential impact radius · As-built status

Charts: Design pressure against wall thickness · one line per design factor · Hoop stress along the route · % SMYS at MAOP against the limit · Steel for this route against line pipe grade

Ready-made scenarios and presets

12 km: farms, village, town · All rural, Class 1 · Same route, Class 1 Division 1 · City edge with multistorey blocks · Built, then 15 years of growth · New school beside a Class 1 line · DN 900 X70 at 100 bar · DN 150 at 20 bar · Compressor discharge at 180 °C

Learning tasks (checked automatically)

1. **Read the baseline.** Record the 12 km route as it opens. Read the length in each location class, the walls used and the steel.

2. **Slide the unit.** Move the pointer along the route. Find a position where the 1-mile unit holds 11–45 buildings (Class 2) and one where it holds 46 or more (Class 3).

3. **Where does the class end?.** Record the route with the cluster rule on and again with it off (Class boundaries). The cluster rule ends Class 2 and 3 at 660 ft past the last building, so less heavy pipe is needed.

4. **Eleven houses.** With the House or Estate tool, add buildings in a Class 1 stretch until it turns into Class 2. Watch F drop from 0.72 to 0.60 and the wall step up.

5. **Concentration of people.** Place a school close to the line in a Class 1 stretch. Table 841.114B treats it as Class 3: F 0.50 near it.

6. **Road crossing.** Add a road crossing in a Class 1 stretch. An uncased crossing of a hard-surface road needs F 0.60 for a short length: find the local wall step.

7. **Buy a better steel.** Record the same route in two grades. Use the grade chart: the steel falls with SMYS until the handling D/t limit takes over.

8. **Too thin to handle.** Find a design where the handling D/t limit, not pressure, sets the wall (small pipe, low pressure or high grade). Record it.

9. **Division 1.** Record the rural route with Class 1 Division 2 and again with Division 1. Division 1 saves steel but needs a 1.25 × MAOP water test.

10. **Hot gas.** Raise the design temperature above 121 °C (compressor discharge). Record it and read the temperature derating factor T.

11. **The town grows.** Build the pipeline, then press +5 years until part of the line is no longer commensurate with its location class (Table 854.1(c)). Record it.

12. **Comply by derating.** Reduce the MAOP to comply and record. Note the capacity you gave up.

13. **Comply by replacing.** Instead keep the MAOP: replace the non-compliant pipe with pipe designed for the new class (854.2(d)). Record the length replaced.

14. **Potential impact radius.** Record two designs whose PIR differs by 30 % or more (change diameter or MAOP). Watch the HCA length follow.

29 · Pipeline Hydrostatic Test Bench

Folder: simulators/pipeline-hydrotest/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/pipeline-hydrotest/index.html

Topic: Gas pipeline strength and leak test · ASME B31.8 841.3 and Appendix N · test sections from the elevation profile · pressure–volume plot · yield detection · 2 h hold · temperature against leak · MAOP from the test

What it does: A 24 km line with 380 m of relief: plan test sections so the high point reaches the minimum test pressure (Table 841.322(f)) without the low point passing the hoop limit; fill behind a pig, hold at 80 %, pressurise along a P–V plot built from water compressibility, pipe expansion, trapped air and yielding joints; N5 yield rules (2 × strokes, 0.002 × fill volume); 2 h strength hold (841.321) with temperature drift, sun and cooling against a hidden leak; construction damage that ruptures; accept or reject and read the MAOP the test supports. Clause numbers follow ASME B31.8-2003.

How to use it

1. Plan the sections: the profile shows 24 km of line with 380 m of relief. Click the profile to add or remove section breaks. The lower chart shows whether the high point can reach the minimum test pressure without the low point passing the hoop limit.

2. Fill the selected section, then Pressurise to 80 % and hold to look for big leaks, then Pressurise to test. Watch the P–V plot: a straight line means elastic pipe.

3. Hold for at least 2 hours with the high point above the minimum. Use the clock to speed time up. Top up with Pump to target if the pressure falls.

4. Judge the hold: correct the pressure change for the temperature change, then Accept the test or Reject it for a leak.

5. Record each section. The scenarios load faults to find: air, cold ground, sun, leaks, yielding pipe and construction damage.

Keys: Space run / pause · F fill · T to test · S stop pump · R record · Esc close

Controls

Outside diameter · Wall · Grade (SMYS) · Required MAOP · Location class (Table 841.322(f)) · Maximum hoop at low point · Fill method (N3) · Fill water temperature · Ground temperature at pipe depth · Pipe exposed to the sun · Pressure pump rate · Clock · Section · Pump to (gauge at the head)

Action buttons: Fill section · Pressurise to 80 % · Pressurise to test · Pump to target · Stop pump · Depressurise · Accept test · Reject: leak · Drain and reset section · Pause · Record reading

Readouts

Gauge at the test head · High point · Low point hoop stress · Water pumped · Water temperature · Strength hold (841.321) · Pressure change in the hold · Test result

Charts: Pressure against volume pumped · the straight line, then deviation (Appendix N5) · Hold log · gauge pressure and water temperature against time · Along the section · test pressure at every point against the limits

Ready-made scenarios and presets

24 km, one section, 380 m of relief · Same line in four sections · Filled without a pig · Warm water into cold ground · Exposed pipe in the sun · Section with a test result to judge (A) · Section with a test result to judge (B) · Pump into yield · Construction damage in the line · Class 3 town section, 1.4 × MAOP

Learning tasks (checked automatically)

1. **Fill a section.** Fill the selected section behind a pig. Read the gauge at the test head when it is full: that is the static head between the head and the vented high point.

2. **Plan the test sections.** With one section the low point would pass the hoop limit before the high point reaches the minimum. Click the profile to add breaks until every section passes (the lower chart helps). Record it.

3. **The 80 % hold (N4).** Pressurise to 80 % of the test pressure and hold for at least 30 minutes, looking for big leaks.

4. **The straight line.** Pressurise to test pressure and read the straight-line slope of the P–V plot in litres and strokes per bar. Record it.

5. **Two hours (841.321).** Hold for 2 hours with the high point at or above the minimum test pressure. Top up if it falls.

6. **Air in the line.** Fill a section without a pig (no venting) and compare the strokes per bar with a pigged fill. Record both.

7. **Find yield (N5).** Test the whole line as one section with the hoop limit at 110 %. Watch the P–V plot leave the straight line and stop when the 2 × strokes or 0.2 % volume rule is reached.

8. **Find the leak.** Judge scenarios A and B. One loses water; the other only cools. Correct the pressure change for temperature before you decide.

9. **Not a leak.** Accept a good section whose pressure changed by more than a bar because the water temperature changed.

10. **Sun on the pipe.** Hold a section with exposed pipe through an afternoon and a night. Watch the pressure follow the sun.

11. **The test did its job.** Load "Construction damage", pressurise to test and find where it fails. Drain, repair, retest and accept the section.

12. **Every section passed.** Accept every section of the line correctly and record the MAOP each test supports.

30 · Pipeline Corrosion and Integrity Assessor

Folder: simulators/pipeline-integrity/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/pipeline-integrity/index.html

Topic: Gas pipeline integrity · cathodic protection survey (on and instant-off, IR drop) · Appendix K / NACE RP0169 criteria · coating and soil · stray current · in-line inspection features · ASME B31.8 Appendix L (original B31G) and modified B31G · repairs per 851.44 · MAOP reduction · remaining life

What it does: A 20 km gas line at year 15. CP solved as a leaky resistor ladder (rectifier, coating conductance, soil, insulating flanges) with a DC-railway stray-current discharge: on and instant-off potentials against −0.85 V and 100 mV polarisation, the IR-drop trap in high-resistivity ground, overprotection below −1.2 V. Twenty-two in-line inspection features assessed with Appendix L ($A = 0.893\,L/\sqrt{Dt}$, short and long formulas, 10 % and 80 % limits) with modified B31G and tool tolerance as comparisons; choose accept, sleeve, cut out, patch (≤ 35 ksi) or weld deposit (burn-through table), or reduce the MAOP; run time forward and watch unprotected features fail; remaining life and re-inspection interval. Clauses follow ASME B31.8-2003.

How to use it

1. CP survey: the plot shows the on and instant-off potentials along 20 km. Click it to read a test post. Raise the rectifier current until the whole route meets a criterion, without going below −1.2 V off.

2. Features: the bars under the plot are corrosion features from an in-line inspection, height = depth. Click one, or use the Anomaly list. The readout shows A, the safe pressure P′ and the verdict.

3. Decide: accept and monitor, sleeve, cut out, patch or weld deposit, or reduce the MAOP. The list says whether your choice is valid.

4. Time: +5 years grows every unrepaired feature at its local corrosion rate. Weak CP, poor coating or the railway will turn features into failures.

Keys: Y +5 years · R record · [/] previous / next feature · Esc close

Controls

Outside diameter · Nominal wall · Grade (SMYS) · Location class (design factor) · MAOP · Assessment · Add tool tolerance: +10 % of wall to every depth · Rectifier current · Coating · Drainage bond to the DC railway at 15 km · Recoat the wetland stretch 4–6 km

Action buttons: Accept, monitor · Full encirclement sleeve · Cut out and replace · Plate patch · Weld deposit · Reduce MAOP to the lowest P′ · +5 years · Record reading

Readouts

Test post · Route meeting a CP criterion · Transformer-rectifier · Selected feature (Appendix L) · Features · Remaining life · MAOP · Service record

Charts: Depth against length · the acceptance curve where P′ = MAOP · Selected feature · projected depth and safe pressure against years · Corrosion rate at a coating holiday along the route

Ready-made scenarios and presets

Year 15: first in-line inspection · CP current too low · High-resistivity ground: the IR-drop trap · Too much current · DC railway interference · Ten years of doing nothing · Old Grade B line (patch allowed) · New FBE-coated line

Learning tasks (checked automatically)

1. **Read a test post.** Click the potential plot to read a test post: on, instant-off, native, polarisation and IR drop.

2. **The IR-drop trap.** Find a test post where the on potential is more negative than −0.85 V but the pipe is not protected (instant-off above −0.85 V and less than 100 mV polarisation). Look in the high-resistivity ground.

3. **Protect the whole line.** Raise the rectifier current until at least 99 % of the route meets a criterion, without any off potential below −1.20 V. Record it.

4. **Too much.** Push the current until the off potential near the drain point is below −1.20 V. Read why that is a problem.

5. **Stray current.** Find where the DC railway current leaves the pipe (potentials swing positive at 15 km). Install the drainage bond and record the difference.

6. **Short and long.** Select a short deep feature ($A \leq 4$) and a long shallow one ($A > 4$). Compare which formula Appendix L uses and why the long one fails at a shallower depth.

7. **Tool tolerance.** Record the assessment with and without +10 % depth tolerance. Count how many more features need action.

8. **Original against modified.** Record the same year with Appendix L and with modified B31G. Compare the lowest P' and the number of features to act on.

9. **Deal with every feature.** Choose a valid action for every feature that needs one (none open, no invalid choices) and record. Over 80 % deep: take it out.

10. **Or reduce the pressure.** Instead of repairing, reduce the MAOP to the lowest P'. Which feature could you still not leave in service?

11. **Ten years of doing nothing.** From the year-15 baseline, leave CP at 1 A and repair nothing. Run +5 years twice. Record the failures and where they happened.

12. **When a patch is allowed.** On the old Grade B line (35 ksi), repair a failing feature with a plate patch. Try the same on X52: why is it not valid?

31 · Pressure Reduction Station Simulator

Folder: simulators/pipeline-overpressure/ · open index.html · online: https://chrispienaarthree.github.io/mech-engineering-simulators/simulators/pipeline-overpressure/index.html

Topic: Gas pressure control and overpressure protection · ASME B31.8 845 and 841.126 · worker and monitor regulators · slam-shut · relief valve sizing · MAOP + 4 % / + 10 % / 75 % SMYS · sensing lines · Joule–Thomson cooling and freezing · outlet velocity

What it does: A let-down station from a 70 bar line to a distribution lateral or transmission spur: water-bath heater, slam-shut, monitor and worker regulators in series (PI control, series flow solved), relief valve with its isolation valve, downstream line pack and a daily demand profile. Inject a worker failing open or shut, a hidden monitor failure, a severed sensing line, a plant trip, an inlet surge or a slam-shut that will not trip, and see which device catches it against 845.411 (MAOP + 4 % above 72 % SMYS, otherwise MAOP + 10 % or 75 % SMYS), 845.421, 845.34–845.36 and 841.126. Size a relief valve alone and count the gas it vents. Clauses follow ASME B31.8-2003.

How to use it

1. Normal day: the worker regulator holds the outlet at its set point while customer demand rises and falls. The monitor sits wide open, waiting.

2. Set points: worker below monitor, monitor at or below MAOP, relief and slam-shut above them but within the allowed overpressure. The checks under the readouts mark each rule.

3. Inject a failure and watch which device catches it. Repair and reset the slam-shut by hand. Reset the highest reading before each new test.

4. Design: remove the monitor and slam-shut and size a relief valve alone; try the old spur at 80 % SMYS; leave the relief isolated; share the sensing line; let the gas go cold.

Keys: Space run / pause · I inject · X repair · R record · Esc close

Controls

Downstream system · Inlet pressure · Peak demand · Inlet gas temperature · Water-bath heater on (outlet at least 5 °C) · Station outlet pipe · Worker regulator set · Monitor regulator installed · Monitor set · Slam-shut installed · Slam-shut trip · Relief valve set · Relief capacity at full lift · Relief isolation valve (845.34) · Sensing lines (845.36) · Clock · Failure

Action buttons: Inject · Repair and reset slam-shut · Reset highest reading · Pause · Record reading

Readouts

Outlet pressure · Highest since reset · Flow · Outlet gas temperature · Outlet velocity at peak · Relief capacity · Gas vented · Last event

Charts: Outlet pressure against time · set points, MAOP and the allowed overpressure · Flow against time · customer demand, station flow and relief · Relief sizing · flow a failed-open regulator passes against what the relief can vent

Ready-made scenarios and presets

Normal day, worker + monitor + slam-shut · Worker fails open · Hidden monitor failure · Relief valve only: size it · Night lock-up, token relief · Relief left isolated · Shared sensing line · Cold gas, heater off · Old spur at 80 % SMYS · Set points in the wrong order

Learning tasks (checked automatically)

1. **A normal day.** Run through a morning and evening peak. Watch the worker open and close while the outlet holds its set point and the monitor waits wide open. Record a reading.

2. **Set the ladder.** Set worker, monitor, relief and slam-shut so that checks C1 to C4 all pass. Record it.

3. **The monitor takes over.** Inject "Worker fails open" with the monitor in service. The outlet settles at the monitor set point, at or below MAOP (845.421).

4. **The hidden failure.** Inject "Monitor stuck open" (nothing happens), then "Worker fails open". What stops it? Why must monitors be checked (845.421)?

5. **Relief valve only.** Remove the monitor and slam-shut. Size the relief so a wide-open worker stays within the limit (use the sizing chart). Inject the failure and record the gas vented.

6. **Night lock-up.** Inject "Demand to zero". Seat leakage creeps the pressure up; a small token relief deals with it before the slam-shut trips.

7. **Relief left isolated.** Close the relief isolation valve and repeat a failure without a monitor or slam-shut. Record what 845.34 is there to prevent.

8. **One line, two regulators.** With one shared sensing line, sever it: both regulators open. Repeat with separate lines (845.36(b)).

9. **Cold gas.** Let the gas go cold with the heater off until the worker pilot freezes. Then switch the heater on (841.126(c)).

10. **Size the outlet.** Raise the peak demand to 50 000 Sm^3/h and choose the smallest outlet pipe that keeps the velocity at or below 30 m/s. Record it.

11. **Four per cent.** On the old spur at 80 % SMYS the limit is MAOP + 4 %. Move the relief and slam-shut so all nine checks pass. Record it.

12. **Manual reset.** After a slam-shut trip, repair the fault and reset it by hand (845.222(e)). Record the reading.